

ECON6101 Microeconomics for Honours Course Notes

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I use course materials from semester 1, 2025 to make most of the notes and then use 2026's materials to refine the notes.

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Part I

Economic Agents

1 Introduction to Consumer Theory

1.1 Logic

“iff” is used for definitions.

Example 1.1.1. A person x is called rational iff x reasons about their preferences and x chooses the best available alternative. This means x must satisfy both properties listed. The property “being rational” is defined by the right hand side of the iff.

“iff” also used in the logical statement “A iff B” i.e. “A if and only if B” i.e. $A \Leftrightarrow B$.

Definition 1.1.2. • $A \Leftrightarrow B$ defined logically to hold if both $A \Rightarrow B$ and $B \Rightarrow A$ hold.
• $A \Rightarrow B$ is defined as $B \vee \neg A$.

There are two logical equivalence table.

A	B	$\neg A$	$\neg B$	$A \Rightarrow B$ ($B \vee \neg A$)	$\neg B \Rightarrow \neg A$ ($\neg A \vee \neg \neg B$)	$\neg A$	$\neg \neg B$
T	T	F	F	T	T	F	T
T	F	F	T	F	F	F	F
F	T	T	F	T	T	T	T
F	F	T	T	T	T	T	F

Note that the contrapositive here: $A \Rightarrow B$ is equivalent to $\neg B \Rightarrow \neg A$.

A	B	C	$A \Rightarrow B \vee C$	$B \vee C \vee \neg A$	$B \vee C$	$\neg A$	$A \wedge \neg B \Rightarrow C$	$C \vee \neg(A \wedge \neg B)$	$\neg(A \wedge \neg B)$	$(A \wedge \neg B)$	$\neg B$
T	T	T	T	T	T	F	T	T	T	F	F
T	T	F	T	T	T	F	T	T	T	F	F
T	F	T	T	T	T	F	T	T	F	T	T
T	F	F	F	F	F	F	F	F	F	T	T
F	T	T	T	T	T	T	T	T	T	F	F
F	T	F	T	T	T	T	T	T	T	F	F
F	F	T	T	T	T	T	T	T	T	F	T
F	F	F	T	T	F	T	T	T	T	F	T

Note that $A \Rightarrow (B \vee C)$ is equivalent to $(A \wedge \neg B) \Rightarrow C$.

1.2 Decision Problems and Binary Relations

Decision maker chooses from a set A of alternatives. A set mathematically is just a list of things.

Example 1.2.1. $A = \{R\&J, O, U, M\}$ (Romeo and Juliet, Origin of Species, Unbearable Lightness of Being, or Advanced Microeconomic Theory). Pick one and only one from this set.

Economics presumes you can rank alternatives and will pick one that is a best option. Maybe there is more than one best option!

Alternative x is best in A iff $x \in A$ and x is “at least as good” as y for every other $y \in A$. We can use $\succeq$ to mean “is at least as good as”. Then we can write:

Definition 1.2.2. x is best in A iff $x \in A$ and $x \succeq y$ for all $y \in A$.

Note the similarity and difference between $\geq$ and $\succeq$. What do we require of $\geq$ when comparing numbers?

1. $a \geq a$ (reflexivity)
2. Either $a \geq b$ or $b \geq a$ (completeness)
3. If $a \geq b$ and $b \geq c$ then $a \geq c$ (transitivity)
4. If $a \geq b$ and $b \geq a$ then $a = b$ (anti-symmetry)

Because all four properties are satisfied for $\geq$ on $\mathbb{R}$, we say $\geq$ is a “total order”. But more generally, it is an example of a binary relation (no axiom is needed!).

We want to say “bundle a is at least as good as bundle b ”. This is a binary relation on a set of alternatives. It is a set of ordered pairs.

Definition 1.2.3. Let X be a set. A binary relation is a subset of $X \times X$.

Re-writing “ $a \succeq b$ ” as “ $(a, b) \in \succeq$ ” reveals that $\succeq$ is a binary relation on the set of alternatives. This one has an interpretation: $(a, b) \in \succeq$ means “ a is at least as good as b ”. Others may be interpretable as “is the mother of”, “loves”, etc. (Or nothing intuitive.)

Example 1.2.4. Suppose ranking is:

$\{U, O\}$ on top, then $R\&J$, then M .

What is the binary relation here?

Here $A = \{U, O, R\&J, M\}$ and $\succeq \subseteq A \times A$. We have:

- $(U, O) \in \succeq$ and $(O, U) \in \succeq$ (tie at top).
- $(U, R\&J), (O, R\&J) \in \succeq$, and $(R\&J, M) \in \succeq$
- $(U, M), (O, M) \in \succeq$
- $(x, x) \in \succeq \quad \forall x \in A$ (reflexivity)

Note $a \succeq b$ is shorthand for $(a, b) \in \succeq$.

In this example, $\succeq \subset A \times A$ (a strict subset). The missing comparisons comprise the complement, $\succeq^c$. Here, $\succeq^c = \{(M, U), (M, O), (R \& J, U), (R \& J, O), (M, R \& J)\}$.

Remark. Sets use $\{\cdot\}$: order does not matter.

$$\{1, 2, 3\} = \{2, 1, 3\}$$

Relations use ordered pairs (a, b) : order does matter.

$$(1, 2) \neq (2, 1).$$

Example 1.2.5. How would you represent $\geq$ as a binary relation on $\mathbb{R} = (-\infty, \infty)$?

$$\geq = \{(a, b) \in \mathbb{R} \times \mathbb{R} : a \geq b\}.$$

Another possible definition:

Definition 1.2.6. x is best in A iff $x \in A$ and no $y \in A$ is strictly better than x . Here we define strictly better than as:

$$y \succ x \iff (y \succeq x) \text{ and } \neg(x \succeq y).$$

Example 1.2.7. Suppose that the binary relation $\succeq$ is empty but A is not empty.

- What is best under Definition 1.2.2? There is no best!
- Definition 1.2.6? Every element is the best!
- Is either definition stronger? Definition 1.2.2 in this case.

Example 1.2.8. Suppose $A = \{a, b, c\}$ is the set of alternatives. Can you find a binary relation on A that does not have a best element according to Definition 1.2.2?

$$\succeq = \{(a, b)\}.$$

According to Definition 1.2.6?

$$\succeq = \{(a, b), (b, c), (c, d)\}.$$

We will use Definition 1.2.2 within this course.

Definition 1.2.9. We often are interested in whether a binary relation satisfies:

- Reflexive: $a \succeq a$ for all a .
- Transitive: $a \succeq b$ and $b \succeq c$ imply $a \succeq c$.
- Complete: for all a, b , either $a \succeq b$ or $b \succeq a$ (or both).
- Antisymmetric (for orders): if $a \succeq b$ and $b \succeq a$, then $a = b$.

Example 1.2.10 (Antisymmetry Example (Numbers)). For $\geq$ on numbers: If $a \geq b$ and $b \geq a$, then $a = b$.

Did previous examples (for the book example) have antisymmetric? No because the books are not identical.

Did previous examples have this transitivity? Yes!

Remark. Completeness: For any $a, b \in X$, we can compare them

- $a \succeq b$ or $b \succeq a$ (possibly both).
- Allows ties (indifference).

1.3 Utility

Completeness and transitivity are enough to create a ranking (formally, a total preorder). Antisymmetry gives a strict order (no ties).

Recall from ECON3010. Let $\succeq$ be a binary relation on a finite set X . Then, $\succeq$ is complete and transitive if and only if it has a utility representation.

Thus, for finite or countable sets, a ranking is enough to be able to represent preferences with a utility function. For uncountable sets, we will need a little more - see next week's lecture.

Definition 1.3.1. A utility representation is a real-valued function $u : X \rightarrow \mathbb{R}$ such that:

$$x \succeq y \iff u(x) \geq u(y).$$

Example 1.3.2.

$$u(U) = u(O) = 2, \quad u(R \& J) = 1, \quad u(M) = 0.$$

Also Recall from ECON3010. Let X be a consumption set and $\succeq$ a complete and transitive preference relation on it. Then every non-empty and finite subset A of X has a greatest element.

Thus, with Definition 1.2.2, as long as $\succeq$ is complete and transitive on a finite set of alternatives, there will be at least one “best” alternative. Otherwise there MAY not be a best alternative.

- What is an example without completeness or transitivity where there is still a best element?

Let $A = \{a, b, c\}$. Define $\succeq = \{(a, a), (b, b), (c, c), (a, b), (a, c)\}$.

- What is an example of a complete and transitive relation on a non-finite set without a best element?

Take $X = \mathbb{R}$. Let the binary relation be the usual “ $\geq$ ”: $\succeq = \{(x, y) \in \mathbb{R}^2 : x \geq y\}$.

2 Preferences and Utility

2.1 Math Review

Definition 2.1.1. A set is a collection of objects. Subset: $S \subseteq T$ iff every element of S is in T . Empty set: $\emptyset$.

We need to distinguish three concepts:

- Set: $\{1, 2, 3\} = \{3, 2, 1\}$.
- Ordered pair: $(1, 2) \neq (2, 1)$.
- n -tuple: $(1, 2, 3) \neq (3, 2, 1)$.

The number of the subsets of a finite set with elements n is 2^n .

Example 2.1.2. $\{a, b, c\}$. There are $2^3 = 8$ subsets.

$\mathbb{R} = (-\infty, \infty)$ also has subsets: $\mathbb{R}$, $\emptyset$, $\mathbb{R}_+ = \{x \in \mathbb{R} : x \geq 0\}$, $\mathbb{R}_{++} = \{x \in \mathbb{R} : x > 0\}$, $A = \{x \in \mathbb{R} : 4 \leq x < 5\} = [4, 5)$, and $\mathbb{N} = \{1, 2, 3, \dots\}$...

$\mathbb{R}$ is our starting point for consumer theory, with its standard properties:

- Field axioms: associative and commutative properties of $+$ and $\times$
- Order axioms: e.g. there is a binary relation $\leq$ that you are familiar with
- Least upper bound property: Each non-empty subset of $\mathbb{R}$ that has an upper bound has a least upper bound

Definition 2.1.3. x is an upper bound of $A \subset \mathbb{R}$ iff $x \geq y$ for all $y \in A$. Least upper bound: the smallest such x .

Example 2.1.4. Is 2 an upper bound for $[\sqrt{2}, 5)$? No. 20? Yes. What is the least upper bound? 5. What about $\mathbb{Q} \cap [-\sqrt{2}, \sqrt{2}]$? $\sqrt{2}$.

Suppose we have many goods. First label the goods by $1, 2, \dots, n$ (for example, 1 is bread, 2 is milk, ...). Let the quantity of each good be x_i , and usually assume quantities cannot be negative: $x_i \geq 0$. A consumption bundle (bundle) is the n -tuple that collects all quantities:

$$\mathbf{x} = (x_1, \dots, x_n).$$

Definition 2.1.5. The consumption set is the set of all feasible consumption bundles.

In this course we usually take

$$\mathbb{R}_+^n = \underbrace{\mathbb{R}_+ \times \dots \times \mathbb{R}_+}_{n \text{ times}}, \quad \text{where } \mathbb{R}_+ = \{x \in \mathbb{R} : x \geq 0\}.$$

Example 2.1.6 ($n = 3$). • $(2, 1, 0) \in \mathbb{R}_+^3$ (zero is allowed)

- $(1, 4, \pi) \in \mathbb{R}_+^3$
- $(0, -1, 5) \notin \mathbb{R}_+^3$ (negative is not allowed)

Also:

$$\mathbb{R}_{++}^n = \{\mathbf{x} \in \mathbb{R}^n : x_i > 0 \forall i\}.$$

This is strict positivity: no coordinate can be 0 (“exclude the boundary 0”).

To talk about complements, we must first specify a universe set Ω : everything we discuss lies in Ω . In this course, commonly $\Omega = \mathbb{R}_+$ or $\Omega = \mathbb{R}$.

Definition 2.1.7. The complement of a set S is defined by

$$S^c = \{x \in \Omega : x \notin S\}.$$

Example 2.1.8. $S = (2, 3) \cup [5, 9)$.

- If $\Omega = \mathbb{R}$, then

$$S^c = (-\infty, 2] \cup [3, 5) \cup [9, \infty).$$

- If $\Omega = \mathbb{R}_+$, add the restriction $x \geq 0$:

$$S^c = [0, 2] \cup [3, 5) \cup [9, \infty).$$

We can define either open sets or closed sets first, and then define the other concept using complements. In this course, we use open balls in $\mathbb{R}^n$ as a working definition to build the notions of openness and closedness.

Definition 2.1.9 (Metric (Distance)). A metric $d(\mathbf{x}, \mathbf{y})$ measures the distance between points $\mathbf{x}$ and $\mathbf{y}$. Euclidean metric on $\mathbb{R}^n$:

$$d(\mathbf{x}, \mathbf{y}) = \sqrt{\sum_{i=1}^n (x_i - y_i)^2}.$$

Key defining properties (metric axioms):

- Triangle inequality:

$$d(\mathbf{x}, \mathbf{y}) \leq d(\mathbf{x}, \mathbf{z}) + d(\mathbf{z}, \mathbf{y}).$$

- Nonnegativity: $d(\mathbf{x}, \mathbf{y}) \geq 0$
- Identity of indiscernibles: $d(\mathbf{x}, \mathbf{y}) = 0 \iff \mathbf{x} = \mathbf{y}$
- Symmetry: $d(\mathbf{x}, \mathbf{y}) = d(\mathbf{y}, \mathbf{x})$

Example 2.1.10 (Illustration in $\mathbb{R}^2$). If the coordinate differences between $\mathbf{x}$ and $\mathbf{y}$ are 1

horizontally and 2 vertically, then

$$d(\mathbf{x}, \mathbf{y}) = \sqrt{1^2 + 2^2} = \sqrt{5}.$$

Definition 2.1.11 (Open Balls). Open ball around $\mathbf{x}$ with radius $\varepsilon > 0$:

$$B_\varepsilon(\mathbf{x}) = \{\mathbf{y} \in \mathbb{R}_+^n : d(\mathbf{x}, \mathbf{y}) < \varepsilon\}.$$

This definition implies two important exclusions:

- It excludes the boundary (points with $d(\mathbf{x}, \mathbf{y}) = \varepsilon$ are not included because of “ $<$ ”).
- It also excludes points with negative coordinates, since the ball is defined with $\mathbf{y} \in \mathbb{R}_+^n$.

Example 2.1.12 (Using the Euclidean metric). Distance between $(1, 0)$ and $(4, 5)$:

$$d((1, 0), (4, 5)) = \sqrt{(1 - 4)^2 + (0 - 5)^2} = \sqrt{9 + 25} = \sqrt{34}.$$

Open ball around $(2, 4)$ with $\varepsilon = 1$:

$$B_1(2, 4) = \{\mathbf{y} = (y_1, y_2) \in \mathbb{R}_+^2 : d((2, 4), \mathbf{y}) < 1\}.$$

Equivalently (Euclidean distance),

$$B_1(2, 4) = \{(y_1, y_2) \in \mathbb{R}_+^2 : (y_1 - 2)^2 + (y_2 - 4)^2 < 1\}.$$

Distance between $(1, 1, 1)$ and $(2, 4, 2)$:

$$d((1, 1, 1), (2, 4, 2)) = \sqrt{(1 - 2)^2 + (1 - 4)^2 + (1 - 2)^2} = \sqrt{1 + 9 + 1} = \sqrt{11}.$$

Definition 2.1.13 (Open Set). $A \subseteq \mathbb{R}_+^n$ is open iff for every $\mathbf{x} \in A$ there is $\varepsilon_{\mathbf{x}} > 0$ such that

$$B_{\varepsilon_{\mathbf{x}}}(\mathbf{x}) \subseteq A.$$

Theorem 2.1.14. *Every open ball is open.*

Proof. Let $S = B_\varepsilon(\mathbf{x}^0)$. Take any $\mathbf{x} \in S$. Then $d(\mathbf{x}, \mathbf{x}^0) < \varepsilon$, so define

$$\delta = \varepsilon - d(\mathbf{x}, \mathbf{x}^0) > 0.$$

For any $\mathbf{y} \in B_\delta(\mathbf{x})$, by the triangle inequality,

$$d(\mathbf{y}, \mathbf{x}^0) \leq d(\mathbf{y}, \mathbf{x}) + d(\mathbf{x}, \mathbf{x}^0) < \delta + d(\mathbf{x}, \mathbf{x}^0) = \varepsilon,$$

hence $\mathbf{y} \in B_\varepsilon(\mathbf{x}^0) = S$. Therefore $B_\delta(\mathbf{x}) \subseteq S$, so S is open. □

Definition 2.1.15 (Closed Set). $A \subseteq \mathbb{R}_+^n$ is closed iff its complement is open.

Example 2.1.16. Complement of a small open ball in $\mathbb{R}_+^2$ is closed. In notation,

$$\left(B_\varepsilon(\mathbf{x})\right)^c = \{\mathbf{y} \in \mathbb{R}_+^2 : d(\mathbf{x}, \mathbf{y}) \geq \varepsilon\},$$

which is closed because it is the complement of an open set.

Example 2.1.17. Working in the universe $\mathbb{R}_+^n$:

- $\mathbb{R}_{++}^n$ is open in $\mathbb{R}_+^n$.
- $\emptyset$ is open in $\mathbb{R}_+^n$.
- Both are also closed (they are complements of each other):

$$(\mathbb{R}_{++}^n)^c = \emptyset, \quad \emptyset^c = \mathbb{R}_{++}^n.$$

The following questions highlight that “open/closed” depends on the chosen universe Ω (and complements are taken relative to Ω).

Example 2.1.18. Universe $[1, 2]$: is $[1, 2)$ open? closed?

Open in $[1, 2]$: YES. For any $x \in [1, 2)$, we can find $\varepsilon > 0$ such that $(x - \varepsilon, x + \varepsilon) \cap [1, 2] \subseteq [1, 2)$. (Equivalently, $[1, 2) = [1, 2] \cap (-\infty, 2)$, and $(-\infty, 2)$ is open in $\mathbb{R}$.)

Closed in $[1, 2]$: NO. The complement in the universe is $[1, 2] \setminus [1, 2) = \{2\}$, which is not open in $[1, 2]$. Hence $[1, 2)$ is not closed in $[1, 2]$.

Example 2.1.19. Universe $\mathbb{R}$: is $[1, 2)$ open? closed?

Open in $\mathbb{R}$: NO. (No open ball around 1 stays inside $[1, 2)$.)

Closed in $\mathbb{R}$: NO. (The point 2 is a limit point but is not included; equivalently, the complement is not open.)

Example 2.1.20. Open ball around the boundary point $\mathbf{x} = (0, 4)$ in $\mathbb{R}_+^2$:

In the universe $\mathbb{R}_+^2$, the open ball is the usual Euclidean ball intersected with $\mathbb{R}_+^2$:

$$B_\varepsilon(\mathbf{x}) = \{\mathbf{y} \in \mathbb{R}_+^2 : d(\mathbf{x}, \mathbf{y}) < \varepsilon\} = \{(y_1, y_2) \in \mathbb{R}_+^2 : y_1^2 + (y_2 - 4)^2 < \varepsilon^2\}.$$

Geometrically, it looks like a “half-disk” (the part with $y_1 \geq 0$) because points with negative y_1 are excluded by the universe $\mathbb{R}_+^2$.

Theorem 2.1.21. $\emptyset$ and $\mathbb{R}^n$ are open.

Proof. $\mathbb{R}^n$: for any $\mathbf{x} \in \mathbb{R}^n$, any ball $B_\varepsilon(\mathbf{x}) \subseteq \mathbb{R}^n$. $\emptyset$: vacuously open (there is no $\mathbf{x} \in \emptyset$ to check). $\square$

Theorem 2.1.22. *Union of any family of open sets is open.*

Proof. If $\mathbf{x} \in \bigcup_{\alpha} A_{\alpha}$, then $\mathbf{x} \in A_{\alpha_0}$ for some α_0 . Since A_{α_0} is open, $\exists \varepsilon > 0$ with $B_{\varepsilon}(\mathbf{x}) \subseteq A_{\alpha_0} \subseteq \bigcup_{\alpha} A_{\alpha}$. $\square$

Theorem 2.1.23. *Intersection of finitely many open sets is open.*

Proof. If $\mathbf{x} \in \bigcap_{i=1}^m A_i$, then for each i there exists $\varepsilon_i > 0$ with $B_{\varepsilon_i}(\mathbf{x}) \subseteq A_i$. Let $\varepsilon = \min\{\varepsilon_1, \dots, \varepsilon_m\}$. Then $B_{\varepsilon}(\mathbf{x}) \subseteq A_i$ for all i , hence $B_{\varepsilon}(\mathbf{x}) \subseteq \bigcap_{i=1}^m A_i$. $\square$

Example 2.1.24 (why “finite” matters). Consider $U_k = (-1/k, 1/k)$. Let $U_k = (-1/k, 1/k)$, each U_k is open in $\mathbb{R}$. But

$$\bigcap_{k=1}^{\infty} U_k = \{0\},$$

and $\{0\}$ is not open in $\mathbb{R}$.

This shows: an infinite intersection of open sets need not be open, which is why the slide states “intersection of finitely many open sets is open.”

To relate unions/intersections with complements, we use De Morgan’s laws:

Definition 2.1.25 (De Morgan’s Laws).

$$\left(\bigcup_i A_i \right)^c = \bigcap_i A_i^c, \quad \left(\bigcap_i A_i \right)^c = \bigcup_i A_i^c.$$

These identities help transfer “open-set” properties to “closed-set” properties via complements. In particular:

- Closed sets: $\emptyset, \mathbb{R}^n$ are closed.
- Finite unions of closed sets are closed.
- Arbitrary intersections of closed sets are closed.

Definition 2.1.26. A set A is bounded if it is contained inside some open ball.

Theorem 2.1.27. *In Euclidean spaces (like $\mathbb{R}_+^n$), a set is compact iff it is closed and bounded.*

Definition 2.1.28 (ε - δ Continuity). Let $f : D \rightarrow \mathbb{R}$. The function f is continuous at $\mathbf{x}$ iff for all $\varepsilon > 0$ there exists $\delta > 0$ s.t.

$$d(\mathbf{x}, \mathbf{y}) < \delta \Rightarrow |f(\mathbf{x}) - f(\mathbf{y})| < \varepsilon.$$

Definition 2.1.29 (Continuity Equivalent Definition). To connect continuity with open sets, let $D \subseteq \mathbb{R}_+^n$. A function $f : D \rightarrow \mathbb{R}$ is continuous iff the inverse image

$$f^{-1}(B) = \{\mathbf{x} \in D : f(\mathbf{x}) \in B\}$$

of each open ball B in the range $\mathbb{R}$ is also open in the domain D .

Definition 2.1.30 (Convex Set). A set $A \subseteq \mathbb{R}_+^n$ is convex iff for all $\mathbf{x}, \mathbf{y} \in A$ and $\alpha \in [0, 1]$,

$$\alpha \mathbf{x} + (1 - \alpha) \mathbf{y} \in A.$$

2.2 Preferences

Let the consumption set be $\mathbb{R}_+^n$. A preference relation is a subset

$$\succeq \subseteq \mathbb{R}_+^n \times \mathbb{R}_+^n.$$

$(\mathbf{x}, \mathbf{y}) \in \succeq$ is written as $\mathbf{x} \succeq \mathbf{y}$. Meaning: bundle $\mathbf{x}$ is at least as good as bundle $\mathbf{y}$.

Example 2.2.1. Four canonical examples:

1. $\succeq_1 = \mathbb{R}_+^n \times \mathbb{R}_+^n$: everything is at least as good as everything.
2. $\succeq_2 = \emptyset$: nothing is comparable (even to itself).
3. $\succeq_3 = \{(\mathbf{x}, \mathbf{y}) : x_i \geq y_i \ \forall i\}$: coordinate-wise order.
4. $\succeq_4 = \{(\mathbf{x}, \mathbf{y}) : \prod_{i=1}^n x_i \geq \prod_{i=1}^n y_i\}$: product index order.

In $\mathbb{R}_+^2$, let $\mathbf{x} = (1, 2)$ and $\mathbf{y} = (2, 1)$.

- Under $\succeq_1$: $\mathbf{x} \succeq \mathbf{y}$ and $\mathbf{y} \succeq \mathbf{x}$.
- Under $\succeq_3$: $\mathbf{x}$ and $\mathbf{y}$ are incomparable.
- Under $\succeq_4$: compare products:

$$(1)(2) = 2, \quad (2)(1) = 2 \Rightarrow \mathbf{x} \succeq_4 \mathbf{y} \text{ and } \mathbf{y} \succeq_4 \mathbf{x}$$

(i.e., they are tied / indifferent under the product index order).

We define “best” using a weak relation:

Definition 2.2.2. $\mathbf{x} \succeq \mathbf{y}$: $\mathbf{x}$ is at least as good as $\mathbf{y}$.

This choice is convenient because: It captures both strict rankings and ties. Verbal interpretation: “better”, “worse”, or “about the same”.

Starting from the weak preference relation $\succeq$, we define two derived relations.

Definition 2.2.3. Strict preference:

$$\mathbf{x} \succ \mathbf{y} \iff \mathbf{x} \succeq \mathbf{y} \text{ and not}(\mathbf{y} \succeq \mathbf{x}).$$

Indifference:

$$\mathbf{x} \sim \mathbf{y} \iff \mathbf{x} \succeq \mathbf{y} \text{ and } \mathbf{y} \succeq \mathbf{x}.$$

(Both directions hold $\Rightarrow \mathbf{x} \sim \mathbf{y}$.)

What Happens in the Four Examples? Applying these definitions to the four canonical relations:

Relation	Indifference $\sim$	Strict preference $\succ$
$\succsim_1$	all pairs	empty
$\succsim_2$	empty	empty
$\succsim_3$	only identical bundles	coordinate-wise $\geq$ with at least one strict
$\succsim_4$	equal products	higher product

Definition 2.2.4 (Lexicographic Preferences). In $\mathbb{R}_+^2$, define lexicographic weak preference by:

$$\mathbf{x} \succeq_L \mathbf{y} \iff (x_1 > y_1) \text{ or } (x_1 = y_1 \text{ and } x_2 \geq y_2).$$

This means: compare the first component first; use the second component only when the first components tie.

Example 2.2.5. 1. $(1, 2)$ vs $(2, 1)$: the second bundle is strictly preferred (larger first component).

2. $(1, 2)$ vs $(1, 1)$: first components tie, so compare second; $(1, 2)$ is preferred.

3. $(1, 2)$ vs $(1, 2)$: indifferent.

From $\succeq_L$, the derived relations are:

- Indifference:

$$\mathbf{x} \sim_L \mathbf{y} \iff \mathbf{x} = \mathbf{y}$$

(only identical bundles are indifferent).

- Strict preference:

$$\mathbf{x} \succ_L \mathbf{y} \iff x_1 > y_1 \text{ or } (x_1 = y_1 \text{ and } x_2 > y_2).$$

- Geometrically, “indifference curves” collapse to single points.

There are as many possible preferences as there are subsets of $\mathbb{R}_+^n \times \mathbb{R}_+^n$. We restrict attention to “reasonable” preferences that satisfy some axioms.

Axiom 2.2.6 (Complete preferences). A preference relation $\succeq$ on $\mathbb{R}_+^n$ is complete if:

$$\forall \mathbf{x}, \mathbf{y} \in \mathbb{R}_+^n, \quad \mathbf{x} \succeq \mathbf{y} \text{ or } \mathbf{y} \succeq \mathbf{x} \text{ (or both).}$$

Interpretation: preferences are rich enough to compare any pair. Economic question: is this always behaviorally realistic?

Example 2.2.7 (Which Examples Satisfy It?). • $\succeq_1$: complete (both directions always hold).

- $\succeq_2$: not complete (no comparisons hold).
- $\succeq_3$: not complete (e.g., $(1, 2)$ and $(2, 1)$).
- $\succeq_4$: complete (compare the real numbers $\prod_i x_i$ and $\prod_i y_i$).
- $\succeq_L$: complete.

Axiom 2.2.8 (Transitive preferences). A preference relation $\succeq$ on $\mathbb{R}_+^n$ is transitive if:

$$\forall \mathbf{x}, \mathbf{y}, \mathbf{z} \in \mathbb{R}_+^n, \quad (\mathbf{x} \succeq \mathbf{y} \text{ and } \mathbf{y} \succeq \mathbf{z}) \Rightarrow \mathbf{x} \succeq \mathbf{z}.$$

Interpretation: no preference cycles in ranking logic.

Example 2.2.9. • $\succeq_1$: transitive.

- $\succeq_2$: transitive (vacuously: antecedent never true).
- $\succeq_3$: transitive (coordinate-wise inequalities chain).
- $\succeq_4$: transitive (real-number order of product index).
- $\succeq_L$: transitive.

Theorem 2.2.10 (Transitivity Transfers). *If $\succeq$ is transitive, then:*

1. Indifference $\sim$ is transitive.
2. Strict preference $\succ$ is transitive.
3. If $\mathbf{x} \succeq \mathbf{y}$ and $\mathbf{y} \succeq \mathbf{z}$ with at least one strict, then $\mathbf{x} \succ \mathbf{z}$.

Proof. (1) $\sim$ is transitive

Recall: $\mathbf{x} \sim \mathbf{y} \iff (\mathbf{x} \succeq \mathbf{y} \text{ and } \mathbf{y} \succeq \mathbf{x})$. Assume $\mathbf{x} \sim \mathbf{y}$ and $\mathbf{y} \sim \mathbf{z}$. Then

- $\mathbf{x} \succeq \mathbf{y}$ and $\mathbf{y} \succeq \mathbf{x}$,
- $\mathbf{y} \succeq \mathbf{z}$ and $\mathbf{z} \succeq \mathbf{y}$.

By transitivity of $\succeq$:

- from $\mathbf{x} \succeq \mathbf{y}$ and $\mathbf{y} \succeq \mathbf{z}$, we get $\mathbf{x} \succeq \mathbf{z}$;
- from $\mathbf{z} \succeq \mathbf{y}$ and $\mathbf{y} \succeq \mathbf{x}$, we get $\mathbf{z} \succeq \mathbf{x}$.

Hence $\mathbf{x} \sim \mathbf{z}$. So $\sim$ is transitive.

(2) $\succ$ is transitive (strict transitivity)

Recall: $\mathbf{x} \succ \mathbf{y} \iff (\mathbf{x} \succeq \mathbf{y} \text{ and } \neg(\mathbf{y} \succeq \mathbf{x}))$. To show: $\mathbf{x} \succ \mathbf{y}$ and $\mathbf{y} \succ \mathbf{z} \Rightarrow \mathbf{x} \succ \mathbf{z}$. Assume $\mathbf{x} \succ \mathbf{y}$ and $\mathbf{y} \succ \mathbf{z}$. Then:

- $\mathbf{x} \succeq \mathbf{y}$ and $\neg(\mathbf{y} \succeq \mathbf{x})$,
- $\mathbf{y} \succeq \mathbf{z}$ and $\neg(\mathbf{z} \succeq \mathbf{y})$.

By transitivity of $\succeq$, from $\mathbf{x} \succeq \mathbf{y}$ and $\mathbf{y} \succeq \mathbf{z}$, we get $\mathbf{x} \succeq \mathbf{z}$.

Now show $\neg(\mathbf{z} \succeq \mathbf{x})$ by contradiction. If $\mathbf{z} \succeq \mathbf{x}$, then with $\mathbf{x} \succeq \mathbf{y}$, transitivity gives $\mathbf{z} \succeq \mathbf{y}$, contradicting $\neg(\mathbf{z} \succeq \mathbf{y})$. Therefore $\neg(\mathbf{z} \succeq \mathbf{x})$, so $\mathbf{x} \succ \mathbf{z}$.

(3) If $\mathbf{x} \succeq \mathbf{y}$ and $\mathbf{y} \succeq \mathbf{z}$ with at least one strict, then $\mathbf{x} \succ \mathbf{z}$

Assume $\mathbf{x} \succeq \mathbf{y}$ and $\mathbf{y} \succeq \mathbf{z}$. By transitivity, $\mathbf{x} \succeq \mathbf{z}$. It remains to prove $\neg(\mathbf{z} \succeq \mathbf{x})$. Consider the two possible “strict” cases:

- Case A: $\mathbf{x} \succ \mathbf{y}$ (so $\neg(\mathbf{y} \succeq \mathbf{x})$). Suppose $\mathbf{z} \succeq \mathbf{x}$. Since $\mathbf{y} \succeq \mathbf{z}$, transitivity implies $\mathbf{y} \succeq \mathbf{x}$, contradicting $\neg(\mathbf{y} \succeq \mathbf{x})$. Hence $\neg(\mathbf{z} \succeq \mathbf{x})$, so $\mathbf{x} \succ \mathbf{z}$.
- Case B: $\mathbf{y} \succ \mathbf{z}$ (so $\neg(\mathbf{z} \succeq \mathbf{y})$). Suppose $\mathbf{z} \succeq \mathbf{x}$. Since $\mathbf{x} \succeq \mathbf{y}$, transitivity implies $\mathbf{z} \succeq \mathbf{y}$, contradicting $\neg(\mathbf{z} \succeq \mathbf{y})$. Hence $\neg(\mathbf{z} \succeq \mathbf{x})$, so $\mathbf{x} \succ \mathbf{z}$.

Therefore, in either case, $\mathbf{x} \succ \mathbf{z}$. □

Axioms 2.2.6 and 2.2.8 are enough to have a utility representation if Ω is countable! But for $\mathbb{R}_+^n$ we need:

Axiom 2.2.11 (Continuity). A preference relation $\succeq$ on $\mathbb{R}_+^n$ is continuous iff for each $\mathbf{x} \in \mathbb{R}_+^n$, the sets

$$G(\mathbf{x}) = \{\mathbf{y} \in \mathbb{R}_+^n : \mathbf{y} \succeq \mathbf{x}\}, \quad W(\mathbf{x}) = \{\mathbf{y} \in \mathbb{R}_+^n : \mathbf{x} \succeq \mathbf{y}\}$$

are closed in $\mathbb{R}_+^n$.

- $G(\mathbf{x})$: “at least as good as $\mathbf{x}$ ” set.
- $W(\mathbf{x})$: “no better than $\mathbf{x}$ ” set.

Geometric Meaning of Continuity: Think of an indifference curve through $\mathbf{x}$.

- $G(\mathbf{x})$: region on and above the curve.
- $W(\mathbf{x})$: region on and below the curve.

Continuity asks both regions to be closed.

Example 2.2.12. Classify these relations:

1. $\mathbf{x} \succeq \mathbf{y} \iff x_1 x_2 \geq y_1 y_2$.
2. $\mathbf{x} \succeq \mathbf{y} \iff x_1^2 + x_2 \geq y_1^2 + y_2$.
3. Lexicographic $\succeq_L$.
4. $\mathbf{x} \succeq \mathbf{y} \iff (x_1 - 3)^2 + (x_2 - 4)^2 \geq (y_1 - 3)^2 + (y_2 - 4)^2$.

Which Are Complete, Transitive, Continuous? (on $\mathbb{R}_+^2$)

- (1) Complete: yes. Transitive: yes. Continuous: yes.
- (2) Complete: yes. Transitive: yes. Continuous: yes.

- (3) Complete: yes. Transitive: yes. Continuous: no.
- (4) Complete: yes. Transitive: yes. Continuous: yes.

Theorem 2.2.13 (Open Strict Sets Characterization). *Suppose preferences are complete. Then continuity is equivalent to:*

$$SG(\mathbf{x}) = \{\mathbf{y} \in \mathbb{R}_+^n : \mathbf{y} \succ \mathbf{x}\} \text{ is open,} \quad SW(\mathbf{x}) = \{\mathbf{y} \in \mathbb{R}_+^n : \mathbf{x} \succ \mathbf{y}\} \text{ is open.}$$

Why? Under completeness:

$$SG(\mathbf{x}) = W(\mathbf{x})^c, \quad SW(\mathbf{x}) = G(\mathbf{x})^c.$$

Theorem 2.2.14 (Intermediate-Value Version). *If preferences are complete and continuous, then:*

$$\mathbf{x} \succ \mathbf{y} \succ \mathbf{z} \Rightarrow \exists \alpha \in (0, 1) \text{ such that } \alpha \mathbf{x} + (1 - \alpha) \mathbf{z} \sim \mathbf{y}.$$

Proof. Define

$$A = \{\alpha \in [0, 1] : \mathbf{y} \succ \alpha \mathbf{x} + (1 - \alpha) \mathbf{z}\}.$$

- $0 \in A$ because $\mathbf{y} \succ \mathbf{z}$ and $\alpha = 0$ implies $\alpha \mathbf{x} + (1 - \alpha) \mathbf{z} = \mathbf{z}$.
- $A \subseteq [0, 1]$, so A is bounded above by 1.
- By the least upper bound property, the supremum exists:

$$\alpha^* = \sup A.$$

Goal: prove $0 < \alpha^* < 1$.

- Show $\alpha^* > 0$:

Since $\mathbf{y} \succ \mathbf{z}$, we have $\mathbf{z} \in SW(\mathbf{y})$, where

$$SW(\mathbf{y}) = \{\mathbf{w} : \mathbf{y} \succ \mathbf{w}\}.$$

Under completeness + continuity (Result 2), strict sets like $SW(\mathbf{y})$ are open. Hence there exists a neighborhood of $\mathbf{z}$ (an open ball around $\mathbf{z}$) contained in $SW(\mathbf{y})$. Points on the segment $\alpha \mathbf{x} + (1 - \alpha) \mathbf{z}$ with $\alpha > 0$ sufficiently small are close to $\mathbf{z}$, so they also lie in $SW(\mathbf{y})$. Therefore, for some small $\bar{\alpha} > 0$,

$$\mathbf{y} \succ \bar{\alpha} \mathbf{x} + (1 - \bar{\alpha}) \mathbf{z},$$

i.e., $\bar{\alpha} \in A$. Hence A contains a positive number, so $\alpha^* = \sup A > 0$.

- Show $\alpha^* < 1$:

Since $\mathbf{x} \succ \mathbf{y}$, we have $\mathbf{x} \in SG(\mathbf{y})$, where

$$SG(\mathbf{y}) = \{\mathbf{w} : \mathbf{w} \succ \mathbf{y}\}.$$

Again, under completeness + continuity, $SG(\mathbf{y})$ is open, so there is a neighborhood of $\mathbf{x}$ contained in $SG(\mathbf{y})$. Points on the segment with α sufficiently close to 1 are close to $\mathbf{x}$, hence lie in $SG(\mathbf{y})$,

meaning they satisfy $\mathbf{w} \succ \mathbf{y}$ and therefore cannot satisfy $\mathbf{y} \succ \mathbf{w}$. Thus there exists $\eta > 0$ such that for all $\alpha \in (1 - \eta, 1]$,

$$\alpha \notin A.$$

So A is bounded above by $1 - \eta < 1$, implying $\alpha^* < 1$.

Define

$$\mathbf{c}^* = \alpha^* \mathbf{x} + (1 - \alpha^*) \mathbf{z}.$$

By completeness, exactly one of the following must hold:

$$\mathbf{c}^* \succ \mathbf{y}, \quad \mathbf{y} \succ \mathbf{c}^*, \quad \mathbf{c}^* \sim \mathbf{y}.$$

We rule out the first two using continuity (openness of strict sets) plus the supremum property.

- Rule out $\mathbf{y} \succ \mathbf{c}^*$:

If $\mathbf{y} \succ \mathbf{c}^*$, then $\mathbf{c}^* \in SW(\mathbf{y})$. Since $SW(\mathbf{y})$ is open, there exists a neighborhood of $\mathbf{c}^*$ contained in $SW(\mathbf{y})$. Points on the segment slightly past $\mathbf{c}^*$ toward $\mathbf{x}$ (i.e., with $\alpha > \alpha^*$ but very close) would also lie in $SW(\mathbf{y})$, implying those $\alpha > \alpha^*$ belong to A . That contradicts α^* being an upper bound (indeed the least upper bound) of A . Hence $\mathbf{y} \not\succ \mathbf{c}^*$.

- Rule out $\mathbf{c}^* \succ \mathbf{y}$:

If $\mathbf{c}^* \succ \mathbf{y}$, then $\mathbf{c}^* \in SG(\mathbf{y})$. Since $SG(\mathbf{y})$ is open, there exists a neighborhood of $\mathbf{c}^*$ contained in $SG(\mathbf{y})$. In particular, along the segment near $\mathbf{c}^*$ from the $\mathbf{z}$ -side, there is an interval $(\alpha^* - \delta, \alpha^*)$ such that for all α in that interval,

$$\alpha \mathbf{x} + (1 - \alpha) \mathbf{z} \in SG(\mathbf{y}) \Rightarrow \neg(\mathbf{y} \succ \alpha \mathbf{x} + (1 - \alpha) \mathbf{z}),$$

so no $\alpha \in (\alpha^* - \delta, \alpha^*)$ can lie in A . But this contradicts the definition of $\alpha^* = \sup A$, which guarantees that A has elements arbitrarily close to α^* from below. Hence $\mathbf{c}^* \not\succ \mathbf{y}$.

Therefore the only remaining case is:

$$\mathbf{c}^* \sim \mathbf{y}.$$

□

This is a strong statement: an indifference point appears on the segment from $\mathbf{z}$ to $\mathbf{x}$. This motivates continuity as a no-jumps assumption.

Why “Intermediate Value”? It compares with the Intermediate Value Theorem (IVT) for continuous real functions. If u represents preferences and is continuous, then

$$\mathbf{x} \succ \mathbf{y} \succ \mathbf{z} \Rightarrow u(\mathbf{x}) > u(\mathbf{y}) > u(\mathbf{z}).$$

IVT then gives a point on the segment $[\mathbf{z}, \mathbf{x}]$ with utility $u(\mathbf{y})$. (Graphically: utility as a function of the segment parameter crosses the level $u(\mathbf{y})$.)

2.3 Utility

Theorem 2.3.1. *If preferences can be represented with a continuous utility function, then those preferences satisfy Axiom 2.2.11 (continuity).*

Remark. It can help us determine whether preferences are continuous.

Proof. Let $u : \mathbb{R}_+^n \rightarrow \mathbb{R}$ represent $\succeq$, i.e.

$$\mathbf{x} \succeq \mathbf{y} \iff u(\mathbf{x}) \geq u(\mathbf{y}).$$

Fix $\mathbf{x}$. Then the upper and lower contour sets become:

$$G(\mathbf{x}) = \{\mathbf{y} : \mathbf{y} \succeq \mathbf{x}\} = \{\mathbf{y} : u(\mathbf{y}) \geq u(\mathbf{x})\} = u^{-1}([u(\mathbf{x}), \infty)),$$

$$W(\mathbf{x}) = \{\mathbf{y} : \mathbf{x} \succeq \mathbf{y}\} = \{\mathbf{y} : u(\mathbf{y}) \leq u(\mathbf{x})\} = u^{-1}((-\infty, u(\mathbf{x})]).$$

The sets $[u(\mathbf{x}), \infty)$ and $(-\infty, u(\mathbf{x})]$ are closed in $\mathbb{R}$. If u is continuous, the preimage of a closed set is closed. Hence $G(\mathbf{x})$ and $W(\mathbf{x})$ are closed for every $\mathbf{x}$, which is exactly Axiom 2.2.11 (closed-set form). $\square$

Theorem 2.3.2. *If a preference relation $\succeq$ is complete, then $u : \mathbb{R}_+^n \rightarrow \mathbb{R}$ is a utility representation of $\succeq$ iff for all bundles $\mathbf{x}, \mathbf{y} \in \mathbb{R}_+^n$:*

1. *If $\mathbf{x} \succ \mathbf{y}$ then $u(\mathbf{x}) > u(\mathbf{y})$.*
2. *If $u(\mathbf{x}) > u(\mathbf{y})$ then $\mathbf{x} \succ \mathbf{y}$.*

Is completeness needed? Yes!

Axiom 2.3.3 (Strong Monotonicity). If $x_i \geq y_i$ for all i and $x_j > y_j$ for some j , then $\mathbf{x} \succ \mathbf{y}$.

In short, “more of at least one good, no less of others” implies strictly better. It is stronger than weak monotonicity formulations.

Theorem 2.3.4 (U-Representation). *If preferences are:*

- *complete,*
- *transitive, and*
- *continuous,*

then a utility representation exists on $\mathbb{R}_+^n$.

Remark. Finite set is not needed!

Proof. If we add in strong monotonicity the proof is a little simpler (see supplementary notes). $\square$

Note: The proof is complicated, but you should be able to show the reverse direction (Continuous U exists $\Rightarrow$ Axioms 1–3 hold).

Proof. Reverse direction: If u represents $\succeq$ in the standard sense $\mathbf{x} \succeq \mathbf{y} \iff u(\mathbf{x}) \geq u(\mathbf{y})$, then:

- Completeness: for any $\mathbf{x}, \mathbf{y}$, either $u(\mathbf{x}) \geq u(\mathbf{y})$ or $u(\mathbf{y}) \geq u(\mathbf{x})$; hence $\mathbf{x} \succeq \mathbf{y}$ or $\mathbf{y} \succeq \mathbf{x}$.
- Transitivity: if $\mathbf{x} \succeq \mathbf{y}$ and $\mathbf{y} \succeq \mathbf{z}$, then $u(\mathbf{x}) \geq u(\mathbf{y}) \geq u(\mathbf{z})$, so $u(\mathbf{x}) \geq u(\mathbf{z})$, hence $\mathbf{x} \succeq \mathbf{z}$.
- Continuity (Axiom 3): if u is continuous, then for each $\mathbf{x}$,

$$G(\mathbf{x}) = \{\mathbf{y} : \mathbf{y} \succeq \mathbf{x}\} = \{\mathbf{y} : u(\mathbf{y}) \geq u(\mathbf{x})\} = u^{-1}([u(\mathbf{x}), \infty))$$

and

$$W(\mathbf{x}) = \{\mathbf{y} : \mathbf{x} \succeq \mathbf{y}\} = \{\mathbf{y} : u(\mathbf{y}) \leq u(\mathbf{x})\} = u^{-1}((-\infty, u(\mathbf{x})])$$

are preimages of closed sets, hence closed.

□

On finite/countable sets: complete + transitive already suffice. If represented by a utility function, preferences are complete and transitive. If represented by a continuous utility, preferences are continuous.

Utility representation is generally not unique. Any strictly increasing transformation preserves preferences. Different representations can simplify different tasks.

Example 2.3.5. Logs can linearize multiplicative forms (same preferences: u vs $\log u$).

In optimization, smooth forms make derivatives and FOCs cleaner.

3 Quasi-concavity and U-max

3.1 Setup

Definition 3.1.1 (Consumer Problem). Find $x^* \in B(p, I) \subseteq \mathbb{R}_+^n$ such that

$$x^* \succeq y \quad \text{for all } y \in B(p, I).$$

This is a strong optimality notion: x^* is at least as good as every feasible bundle. Alternative (weaker) notion: no feasible bundle is strictly better. In this course we use the strong notion under complete preferences.

Definition 3.1.2. Given prices $p = (p_1, \dots, p_n) \in \mathbb{R}_{++}^n$ and income $I \in \mathbb{R}_+$:

- Budget set:

$$B(p, I) = \{x \in \mathbb{R}_+^n : p \cdot x \leq I\}.$$

- Budget line:

$$p \cdot x = I.$$

For 2-good, the budget line is $p_1x_1 + p_2x_2 = I$. In n goods, this is a hyperplane boundary of $B(p, I)$.

To guarantee a solution to utility maximization,

$$\max\{u(x) : x \in B(p, I)\},$$

we use:

Theorem 3.1.3 (Weierstrass Theorem). *A continuous real-valued function on a non-empty compact set attains a maximum (and minimum).*

So we need: continuity of u , and non-empty compact $B(p, I)$.

To support existence results (e.g., via compactness of the feasible set), we verify that the budget set $B(p, \ell)$ is compact when prices are strictly positive.

1. Non-empty: $0 \in B(p, \ell)$ whenever $\ell \geq 0$.
2. Closed: $\{x : p \cdot x \leq \ell\}$ is closed; its intersection with $\mathbb{R}_+^n$ is also closed.
3. Bounded: with $p_i > 0$, each coordinate satisfies

$$x_i \leq \frac{\ell}{p_i}.$$

Hence $B(p, \ell)$ is (with strictly positive prices) a compact triangle-like region (in the 2-good picture).

Recall that a set $A \subseteq \mathbb{R}^n$ is convex if every line segment between two points in A stays in A . Formally, $A \subseteq \mathbb{R}^n$ is convex iff for all $x, y \in A$ and $\alpha \in [0, 1]$,

$$\alpha x + (1 - \alpha)y \in A.$$

Example 3.1.4. • Convex: balls, boxes, half-spaces, budget sets with linear prices.
• Non-convex: donut/ring shapes (with a “hole”), kinked discount sets.

Theorem 3.1.5. $B(p, I)$ is convex.

Proof. If $x, y \in B(p, I)$ and $t \in [0, 1]$, then

$$p \cdot (tx + (1 - t)y) = tp \cdot x + (1 - t)p \cdot y \leq tI + (1 - t)I = I.$$

So $tx + (1 - t)y \in B(p, I)$. □

Non-convex budget sets can occur. Convexity can fail when prices depend on quantity.

Example 3.1.6. Bulk discounts after thresholds. The set may remain closed and bounded, but develop a “dent” (hence become non-convex).

Existence can still hold; however, simple FOC methods may fail.

Theorem 3.1.7 (Weierstrass’s Theorem). *Every continuous real-valued function f on a non-empty compact set $B \subset \mathbb{R}^n$ attains a maximum value on B . That is, there exists some $x \in B$ such that*

$$f(x) \geq f(y) \quad \forall y \in B.$$

This is an existence result, not yet uniqueness. Multiple maximizers are possible without strict curvature assumptions.

Theorem 3.1.8 (Weierstrass's theorem (consumer choice version)). *If preferences admit a continuous utility representation u , and the budget set $B(p, I)$ is non-empty and compact, then an optimal bundle exists.*

Below checks connect directly to the assumptions behind Weierstrass (continuity + non-empty compact feasible set).

1. Is the budget line compact in $\mathbb{R}^n$?

No. In $\mathbb{R}^n$, the budget line $\{x \in \mathbb{R}^n : p \cdot x = I\}$ is a hyperplane, which is closed but unbounded, hence not compact.

2. Can existence fail if the feasible set is open (not closed)?

Yes. A maximum may not be attained on an open set (e.g., maximizing a continuous objective can yield only a supremum on the boundary that is not included).

3. Can existence fail if the set is unbounded?

Yes. If the feasible set is unbounded and utility can increase without bound along feasible directions, there may be no maximizer (only “larger and larger” feasible values).

4. Can existence fail if utility is discontinuous?

Yes. Even on a compact feasible set, discontinuity can prevent attainment of a maximum (e.g., the function may approach arbitrarily high values near a point without achieving them).

3.2 Convex Preferences and Quasi-Convavity

Building on the notion of convex sets, we define convexity for preferences via upper contour sets.

Definition 3.2.1. Preferences $\succeq$ are convex if for every $x \in \mathbb{R}_+^n$, the upper contour set

$$G(x) = \{y \in \mathbb{R}_+^n : y \succeq x\}$$

is convex.

Economic interpretation: averages are at least as good as extremes.

Remark. For complete and transitive preferences, convexity is equivalent to:

$$x \succeq y \Rightarrow tx + (1 - t)y \succeq y, \quad \forall t \in [0, 1].$$

In words, if x is at least as good as y , then every average with y is also at least as good as y .

Strengthening convexity leads to strict convexity.

Definition 3.2.2. $\succeq$ is strictly convex if for all distinct x, y and all $t \in (0, 1)$:

$$x \succeq y \Rightarrow tx + (1 - t)y \succ y.$$

Economic interpretation: averages are strictly better than extremes.

This diagram illustrates the difference between convex (weak) and strictly convex behavior using two bundles x and y in (x_1, x_2) -space.

- Convex (weak): the line segment between x and y is drawn as a straight segment, indicating that taking averages (mixtures) of two bundles need only be no worse than the extremes; the “average” can lie on the same flat segment (so not strictly better).
- Strictly convex: the curve is bowed, and the mixture point $tx + (1 - t)y$ is shown strictly above the curve through x and y , indicating that averages are strictly better than extremes (for $t \in (0, 1)$).

Remark. With diminishing MRS and "more is better", we can say the preference is convex, but not the other way around.

We now move to Quasi-Concavity.

Careful: convexity of a function is not the same as convexity of preferences.

Definition 3.2.3 (Function concavity). A real-valued function is concave if for any x, y on its domain,

$$f(tx + (1 - t)y) \geq tf(x) + (1 - t)f(y) \quad \forall t \in [0, 1].$$

The region “below” the graph is convex for a concave function; the set “above” is convex for a convex function.

Definition 3.2.4 (Quasi-Concavity of Utility). $u : D \rightarrow \mathbb{R}$, where D is a convex subset of $\mathbb{R}_+^n$ is quasi-concave if for all x, y and $t \in [0, 1]$,

$$u(x) \geq u(y) \Rightarrow u(tx + (1 - t)y) \geq u(y).$$

Remark. Equivalent: all upper contour sets $\{z : u(z) \geq c\}$ are convex.

Definition 3.2.5 (Strict Quasi-Concavity). u is strictly quasi-concave if for all distinct x, y and all $t \in (0, 1)$,

$$u(x) \geq u(y) \Rightarrow u(tx + (1 - t)y) > u(y).$$

Implication: guarantees uniqueness of the maximizer on convex feasible sets.

The relationship between concavity and quasi-concavity:

- Every concave function is quasi-concave.
- Every strictly concave function is strictly quasi-concave.
- The converse is false: quasi-concavity is weaker.

Example 3.2.6. $u(x) = x^2$ on $\mathbb{R}_+$.

On $\mathbb{R}_+$, x^2 is strictly increasing. **Any strictly increasing 1D function is quasi-concave on its domain** (upper contour sets are rays/intervals). Therefore $u(x) = x^2$ is quasi-concave (indeed strictly quasi-concave) on $\mathbb{R}_+$, despite being convex as a function.

Example 3.2.7. $u(x_1, x_2) = x_1^2 + x_2$. Is it concave? No.

Is it quasi-concave on $\mathbb{R}_+^2$? No. (Even though both x_1^2 and x_2 are quasi-concave.)

Upper contour sets: for any c ,

$$\{(x_1, x_2) \in \mathbb{R}_+^2 : x_1^2 + x_2 \geq c\} = \{(x_1, x_2) \in \mathbb{R}_+^2 : x_2 \geq c - x_1^2\}.$$

This is the region above the curve $x_2 = c - x_1^2$ (restricted to $x_1 \geq 0, x_2 \geq 0$).

Direct non-convexity check (counterexample): take $c = 1$.

- $A = (0, 1)$ satisfies $0^2 + 1 = 1 \geq 1$.
- $B = (1, 0)$ satisfies $1^2 + 0 = 1 \geq 1$.
- Midpoint $M = (0.5, 0.5)$ gives $0.5^2 + 0.5 = 0.25 + 0.5 = 0.75 < 1$, so $M \notin$ the upper contour set.

Hence the upper contour set is not convex, so u is not quasi-concave on $\mathbb{R}_+^2$.

Caution:

- Sum of concave functions is concave.
- Sum of quasi-concave functions need not be quasi-concave.
- Always verify quasi-concavity directly or via recognized sufficient conditions.

3.3 Lagrange Method

Assume u is continuously differentiable, preferences are strictly monotonic, and u is quasi-concave.

Consider the utility maximization problem:

$$\max_{x \in \mathbb{R}_+^n} u(x) \quad \text{s.t.} \quad p \cdot x \leq I.$$

Use a Lagrangian with multiplier $\lambda \geq 0$:

$$\mathcal{L} = u(x) - \lambda(p \cdot x - I).$$

For interior candidates (all $x_i > 0$), the FOCs are:

$$\frac{\partial \mathcal{L}}{\partial x_i} = \frac{\partial u(x^*)}{\partial x_i} - \lambda^* p_i = 0 \quad (i = 1, \dots, n),$$

$$\frac{\partial \mathcal{L}}{\partial \lambda} = p \cdot x^* - I = 0.$$

Theorem 3.3.1 (Sufficiency of the Lagrangian Method). *If u is continuously differentiable, strictly monotonic, and quasi-concave, and (x^*, λ^*) solves the first-order system with $(p, I) \in \mathbb{R}_{++}^n \times \mathbb{R}_+$, then x^* solves the consumer problem.*

Remark. Preferences are strictly (strongly) monotonic iff their utility representations are strictly (strongly) increasing.

Remark. Under strict monotonicity, the budget binds. Under quasi-concavity, the FOCs are sufficient (not just necessary).

From the FOCs at an interior optimum x^* ,

$$\frac{\partial u(x^*)/\partial x_i}{p_i} = \frac{\partial u(x^*)/\partial x_j}{p_j} = \lambda^* \quad \forall i, j. \quad (\text{Bang-for-Buck Equalization})$$

Interpretation: the marginal utility per dollar is the same across all goods. Equivalent two-good statement:

$$MRS_{12}(x^*) = \frac{p_1}{p_2}.$$

- Interior optimum: tangency between the indifference curve (IC) and the budget line at x^* .
- Strict monotonicity pushes the optimum to the boundary, i.e. the budget is exhausted (the constraint binds).

Theorem 3.3.2. *If u is strictly quasi-concave, there is at most one solution.*

- Existence comes from continuity + compact budget set (Weierstrass).
- Uniqueness comes from strict quasi-concavity + convex feasible set.

Example 3.3.3. $u(x, y) = x + \sqrt{y}$, prices $(p_x, p_y) = (2, 4)$, income $I = 8$.

Set up the Lagrangian:

$$\mathcal{L}(x, y, \lambda) = x + \sqrt{y} + \lambda(8 - 2x - 4y).$$

FOCs:

$$\frac{\partial \mathcal{L}}{\partial x} = 1 - 2\lambda = 0, \quad \frac{\partial \mathcal{L}}{\partial y} = \frac{1}{2\sqrt{y}} - 4\lambda = 0, \quad \frac{\partial \mathcal{L}}{\partial \lambda} = 8 - 2x - 4y = 0.$$

Solve the FOCs (interior candidate):

- From $1 - 2\lambda = 0$: $\lambda^* = \frac{1}{2}$.
- From $\frac{1}{2\sqrt{y}} - 4\lambda = 0$: $\frac{1}{2\sqrt{y^*}} = 4 \cdot \frac{1}{2} = 2 \Rightarrow \sqrt{y^*} = \frac{1}{4} \Rightarrow y^* = \frac{1}{16}$.
- From $8 - 2x - 4y = 0$: $2x + 4y = 8 \Rightarrow x + 2y = 4 \Rightarrow x^* = 4 - 2 \cdot \frac{1}{16} = 4 - \frac{1}{8} = \frac{31}{8}$.

Check feasibility and assumptions (as the slide instructs):

- $x^* = \frac{31}{8} > 0$, $y^* = \frac{1}{16} > 0 \Rightarrow$ the candidate is interior and satisfies $x, y \geq 0$.
- Budget binds (also shown by the third FOC): $2x^* + 4y^* = 8$.
- u is continuously differentiable on $\{y > 0\}$ and the optimum has $y^* > 0$.
- u is strictly increasing in both goods on $\mathbb{R}_+^2$, and u is concave (hence quasi-concave), so the FOC method applies here.

The Lagrangian technique can fail (or require modifications) under:

- Non-differentiability (kinks).
- Corner solutions (some $x_i^* = 0$).
- Non-convex feasible sets.

- Non-continuous preferences/utility.

Practical advice: be slow, methodical, and geometric: draw ICs and the feasible set, then compare candidate bundles directly.

Perfect complements and perfect substitutes are key preference classes that often generate corner solutions, making interior FOCs uninformative.

- Perfect complements: kinked indifference curves (kinked ICs). Not differentiable.
- Perfect substitutes: linear indifference curves (linear ICs). Not strictly quasi-concave.

Both can place optima at corners, where interior FOCs are not informative.

Example 3.3.4. $u(x, y) = \min\{x, y\} + x + y$. This utility combines different structural components:

- It combines complementarity and substitutability components.
- Its ICs are piecewise linear with kinks and changing slopes.

Recall the Lexicographic Preferences on $\mathbb{R}_+^2$:

$$x \succeq y \text{ if either } x_1 > y_1, \text{ or } (x_1 = y_1 \text{ and } x_2 \geq y_2).$$

Key properties:

- Complete and transitive, but not continuous.
- No continuous utility representation on $\mathbb{R}_+^2$.

Thus, standard smooth optimization tools do not apply here.

Note that “more x_1 ” dominates any change in x_2 . Optimal choice under a standard budget: spend to maximize x_1 first (the optimum lies at the feasible point with the largest x_1 on the budget).

3.4 Summary

- Existence: continuity + non-empty compact budget set.
- The budget set is compact and convex under standard assumptions ($p \in \mathbb{R}_{++}^n$, $I \geq 0$).
- Convex preferences $\iff$ quasi-concave utility representation.
- Strict quasi-concavity gives uniqueness.
- Lagrangian FOCs deliver bang-for-buck equalization and budget exhaustion in smooth interior cases.
- When assumptions fail (kinks, corners, non-convexities, lexicographic), use direct geometric/methodical analysis.

Checklist for Solving U-Max Problems

1. Write the feasible set clearly (including non-negativity).
2. Check continuity and compactness for existence.

3. Check monotonicity and (strict) quasi-concavity.
4. Use FOCs only when smooth interior assumptions are justified.
5. If not, inspect corners/kinks and compare utility directly.

4 Demand and Consumer Welfare

4.1 Marshallian Demand

Recall from last section. Utility maximization problem: Given prices $\mathbf{p}$ and income I , the consumer solves

$$\max\{u(\mathbf{x}) : \mathbf{x} \in B(\mathbf{p}, I)\}, \quad B(\mathbf{p}, I) = \{\mathbf{x} \in \mathbb{R}_+^n : \mathbf{p} \cdot \mathbf{x} \leq I\}.$$

- Under continuity and compactness, a solution exists.
- Under strict quasi-concavity, the solution is unique.

Because the solution to the utility-maximization problem may or may not be unique, we define

Definition 4.1.1. Define the demand correspondence as the set of optimal bundles:

$$X(\mathbf{p}, I) = \{\mathbf{x} \in B(\mathbf{p}, I) : \mathbf{x} \succeq \mathbf{y} \text{ for all } \mathbf{y} \in B(\mathbf{p}, I)\}.$$

This is called Marshallian demand correspondence.

It maps $(\mathbf{p}, I) \in \mathbb{R}_{++}^n \times \mathbb{R}_+$ into a set of optimal bundles. This set can have one point, many points, or (without assumptions) can fail to be well-defined.

To move from a correspondence to a function:

- If $X(\mathbf{p}, I)$ is a singleton, write $\mathbf{x}(\mathbf{p}, I)$ (the Marshallian demand function).
- If $X(\mathbf{p}, I)$ is not a singleton, keep the set notation $X(\mathbf{p}, I)$.

This distinction is important in examples like perfect substitutes with tied prices, where multiple optimal bundles can occur.

Example 4.1.2 (Perfect Substitutes). Let $u(x_1, x_2) = x_1 + x_2$. Then the demand correspondence is

$$X(\mathbf{p}, I) = \begin{cases} \{(I/p_1, 0)\}, & p_1 < p_2, \\ \{(0, I/p_2)\}, & p_2 < p_1, \\ \{\mathbf{x} \in \mathbb{R}_+^2 : \mathbf{p} \cdot \mathbf{x} = I\}, & p_1 = p_2. \end{cases}$$

The tie case ($p_1 = p_2$) produces many optimizers on the budget line.

Building on the idea that some preferences generate unique demands while others may not, the following canonical examples summarize when demand is unique:

Example 4.1.3 (Canonical Demand Examples). 1. Cobb–Douglas:

$$u(\mathbf{x}) = \prod_{j=1}^n x_j^{\alpha_j}, \quad \alpha_j > 0.$$

Demands are unique:

$$x_i(\mathbf{p}, I) = \frac{\alpha_i I}{k p_i} \quad \text{where } k = \sum_i \alpha_i.$$

2. Perfect complements:

$$u(\mathbf{x}) = \min\{x_1, \dots, x_n\}.$$

Demands are unique:

$$x_i(\mathbf{p}, I) = \frac{I}{\sum_j p_j}.$$

3. Perfect substitutes:

$$u(\mathbf{x}) = \sum_j x_j.$$

Demands are not unique if there is more than one lowest price.

Definition 4.1.4 (Quasi-Linear Utility). A useful class that often restores uniqueness and simplifies income effects is quasi-linear utility:

$$u(\mathbf{x}) = v(x_1, \dots, x_{n-1}) + x_n.$$

Technically, perfect substitutes fit this definition.

Standard regularity conditions: v is twice differentiable, increasing, concave (often strictly). Then preferences are convex; often strictly convex with strict concavity of v . In this case, demands are unique and demand for x_1 is independent of income if income is sufficiently high.

Definition 4.1.5. For $f : X \subseteq \mathbb{R}^n \rightarrow \mathbb{R}$:

- Increasing:

$$\mathbf{x} \geq \mathbf{y} \Rightarrow f(\mathbf{x}) \geq f(\mathbf{y}).$$

- Strictly increasing:

$$\mathbf{x} > \mathbf{y} \Rightarrow f(\mathbf{x}) > f(\mathbf{y}).$$

- Strongly increasing:

$$\mathbf{x} \geq \mathbf{y} \text{ and } x_j > y_j \text{ for some } j \Rightarrow f(\mathbf{x}) > f(\mathbf{y}).$$

Remark. Strongly increasing function $\subset$ Strictly increasing function $\subset$ Increasing function.

The distinction between strict and strong only matters in multi-dimensional domains.

Definition 4.1.6. For $\succeq$ on $\mathbb{R}_+^n$,

- Strict monotonicity: If $x_i \geq y_i$ for all i then $\mathbf{x} \succeq \mathbf{y}$, and also if $x_i > y_i$ for all i then $\mathbf{x} \succ \mathbf{y}$.

(More of everything is strictly better.)

- Strong monotonicity: If $x_i \geq y_i$ for all i and $x_j > y_j$ for some j , then $\mathbf{x} \succ \mathbf{y}$. (More is strictly better.)

Remark. Axiom 4 from Lecture 2 is strong monotonicity. With complete preferences, it implies strict monotonicity.

Theorem 4.1.7. *If preferences $\succeq$ over $X \subseteq \mathbb{R}^n$ are represented by a utility function u , then $\succeq$ are strictly (strongly) monotonic if and only if u is strictly (strongly) increasing over X .*

Definition 4.1.8 (Homogeneity of degree k). A function $f : X \rightarrow \mathbb{R}$ with $X \subseteq \mathbb{R}^n$ is homogeneous of degree k iff

$$f(t\mathbf{x}) = t^k f(\mathbf{x}) \quad \text{for all } t > 0 \text{ and all } \mathbf{x} \in X.$$

- Degree 1: doubling all arguments doubles the value.
- Degree 0: scale-invariant.
- Example: perfect substitutes/complements are degree 1.

Example 4.1.9 (Cobb–Douglas Homogeneity). Consider Cobb–Douglas utility:

$$u(\mathbf{x}) = \prod_{j=1}^n x_j^{\alpha_j}, \quad \alpha_j > 0 \text{ for all } j = 1, \dots, n.$$

Let

$$\alpha_1 + \dots + \alpha_n = k \quad \text{for some } k > 0.$$

For $t > 0$,

$$u(t\mathbf{x}) = \prod_{j=1}^n (tx_j)^{\alpha_j} = \left(\prod_{j=1}^n t^{\alpha_j} \right) \left(\prod_{j=1}^n x_j^{\alpha_j} \right) = t^k u(\mathbf{x}).$$

Hence, Cobb–Douglas utility is homogeneous of degree

$$k = \alpha_1 + \alpha_2 + \dots + \alpha_n.$$

Macro people like to use this as a production function where

$$\alpha_1 + \alpha_2 + \dots + \alpha_n = 1.$$

Example 4.1.10 (Quasi-linear (non-)homogeneity).

$$u(\mathbf{x}) = \sqrt{x_1} + x_2$$

is not homogeneous of any degree (most functions are not). How to see it is not: fix $t = 4$.

First, $u(1, 1) = 2$ and

$$u(4(\mathbf{1}, \mathbf{1})) = u(4, 4) = \sqrt{4} + 4 = 6 = 3u(1, 1).$$

If it were homogeneous of degree k , then $4^k = 3$.

- However, $u(4, 1) = 3$ and

$$u(4(4, 1)) = u(16, 4) = \sqrt{16} + 4 = 8 = \frac{8}{3}u(4, 1).$$

This suggests that if it were homogeneous of degree k , then $4^k = \frac{8}{3}$.

Since $3 \neq \frac{8}{3}$, this utility function cannot be homogeneous of any degree k .

Lemma 4.1.11. Assume preferences on $\mathbb{R}_+^n$ are represented by a utility function u that is continuous and strongly monotonic. Then for all $(\mathbf{p}, I) \in \mathbb{R}_{++}^n \times \mathbb{R}_+$, the demand correspondence $X(\mathbf{p}, I)$ satisfies:

1. Non-empty.

2. Budget exhaustion:

$$\mathbf{p} \cdot \mathbf{x} = I \quad \text{for all } \mathbf{x} \in X(\mathbf{p}, I).$$

3. Homogeneous of degree 0 in $(\mathbf{p}, I)$:

$$X(\mathbf{p}, I) = X(t\mathbf{p}, tI), \quad t > 0.$$

4. Continuity under uniqueness: If demand is unique for each $(\mathbf{p}, I)$, then the Marshallian demand function $\mathbf{x}(\mathbf{p}, I)$ is continuous.

- Note: when it is not a singleton set, the demand correspondence is upper semi-continuous / hemi-continuous (not needed for this class).

5. Convexity under quasi-concavity: If u is quasi-concave, then $X(\mathbf{p}, I)$ is a convex set.

6. Uniqueness under strict quasi-concavity: If u is strictly quasi-concave, then $X(\mathbf{p}, I)$ is a singleton (uniqueness).

Why degree-0 homogeneity in $(\mathbf{p}, I)$:

Scaling all prices and income by $t > 0$ leaves the budget set unchanged. Therefore the optimizing bundles are unchanged: there is no money illusion in the model (the budget line for $(\mathbf{p}, I)$ is the same as for $(t\mathbf{p}, tI)$).

4.2 Indirect Utility

Definition 4.2.1 (Indirect Utility Function). The indirect utility function is defined by

$$v(\mathbf{p}, I) = \max\{u(\mathbf{x}) : \mathbf{x} \in \mathbb{R}_+^n, \mathbf{p} \cdot \mathbf{x} \leq I\}.$$

If demand is unique, then

$$v(\mathbf{p}, I) = u(\mathbf{x}(\mathbf{p}, I)).$$

It summarizes welfare level as a function of the market environment $(\mathbf{p}, I)$.

Example 4.2.2 (Cobb–Douglas Indirect Utility). For Cobb–Douglas utility

$$u(\mathbf{x}) = \prod_j x_j^{\alpha_j}, \quad \alpha_j > 0,$$

optimal demand is

$$x_j(\mathbf{p}, I) = \frac{\alpha_j}{k} \frac{I}{p_j} \quad \text{with } k = \sum_k \alpha_k.$$

Plugging back yields

$$v(\mathbf{p}, I) = \prod_j \left(\frac{\alpha_j}{k} \frac{I}{p_j} \right)^{\alpha_j} = \left(\frac{I}{k} \right)^k \prod_j \left(\frac{\alpha_j}{p_j} \right)^{\alpha_j}.$$

Notice: Goods are normal. Income up $\Rightarrow$ demand, utility up. Opposite for prices.

Demands are quite general and allowed to go up, down, or stay the same with prices and income. Recall: substitutes, complements, normal and inferior goods. For indirect utility we will get some general properties regarding prices and income. For that, we need two more definitions.

Definition 4.2.3 (Quasi-Convexity). For $f : \mathbb{R}_+^n \rightarrow \mathbb{R}$, f is quasi-convex iff

$$f(\mathbf{x}) \geq f(\mathbf{y}) \Rightarrow f(t\mathbf{x} + (1-t)\mathbf{y}) \leq f(\mathbf{x}), \quad t \in [0, 1].$$

In words, averages are worse than extremes. In quasi-convexity, the average must be worse than the higher extreme, while in quasi-concavity, the average must be better than the lower extreme.

Theorem 4.2.4. A function f is quasi-convex if and only if $-f$ is quasi-concave.

A quasi-convex function has convex inferior (worse-than) sets.

Theorem 4.2.5. If f is convex, then f is quasi-convex.

Just as quasi-concavity is related to concavity of a function, quasi-convexity is related to convexity of a function, not to convexity of a set.

Example 4.2.6. Linear indifference curves indicate that perfect substitute utility functions are both quasi-convex and quasi-concave.

Example 4.2.7. For $u(x, y) = x^2 + y$, indifference curves $u(x, y) = c$ are

$$y = c - x^2,$$

downward-opening parabolas. The lower contour set $\{(x, y) : x^2 + y \leq c\} = \{(x, y) : y \leq c - x^2\}$ is the region below a concave curve, which is convex, so u is quasi-convex. The upper contour set $\{(x, y) : x^2 + y \geq c\} = \{y \geq c - x^2\}$ is generally not convex, so u is not quasi-concave.

Definition 4.2.8. For $f : X \subseteq \mathbb{R}^n \rightarrow \mathbb{R}$, decreasing is defined as

$$\mathbf{x} \geq \mathbf{y} \Rightarrow f(\mathbf{x}) \leq f(\mathbf{y}).$$

Remark. Analogous definitions for strictly and strongly decreasing functions.

Theorem 4.2.9 (Properties of Indirect Utility). *If u is continuous and strongly monotonic, then the indirect utility $v(\mathbf{p}, I)$ is:*

1. continuous on $\mathbb{R}_{++}^n \times \mathbb{R}_+$,
2. homogeneous of degree 0,
3. strictly increasing in I ,
4. decreasing in $\mathbf{p}$,
5. quasi-convex in $(\mathbf{p}, I)$.

Theorem 4.2.10 (Roy's Identity). *If v is differentiable at $(\mathbf{p}, I)$ and $\partial v / \partial I \neq 0$, then*

$$x_i(\mathbf{p}, I) = - \frac{\partial v(\mathbf{p}, I) / \partial p_i}{\partial v(\mathbf{p}, I) / \partial I}, \quad i = 1, \dots, n.$$

This is convenient to recover Marshallian demand from indirect utility by differentiation.

4.3 Dual Problem: Expenditure Minimization

Definition 4.3.1 (Expenditure Function). Given prices $\mathbf{p}$ and a utility target $\bar{u}$, define the expenditure function as

$$e(\mathbf{p}, \bar{u}) = \min_{\mathbf{x} \in \mathbb{R}_+^n} \{\mathbf{p} \cdot \mathbf{x} : u(\mathbf{x}) \geq \bar{u}\}.$$

The solution $\mathbf{x}^h(\mathbf{p}, \bar{u})$ is the Hicksian (compensated) demand.

It gives the minimal cost to reach the utility target $\bar{u}$ at prices $\mathbf{p}$.

To find $\mathbf{x}^h(\mathbf{p}, \bar{u})$ and $e(\mathbf{p}, \bar{u})$:

- Fix the utility contour $\bar{u}$.
- Slide the expenditure line inward until the first tangency.
- The tangency point gives $\mathbf{x}^h(\mathbf{p}, \bar{u})$ and $e(\mathbf{p}, \bar{u})$, with

$$e(\mathbf{p}, \bar{u}) = \mathbf{p} \cdot \mathbf{x}^h(\mathbf{p}, \bar{u}).$$

Finding the expenditure function: Solving the e-min problem can be done in the differentiable case using Lagrangian-type methods. Under standard assumptions of quasi-concave utility, strong monotonicity, and an interior solution:

- it satisfies Bang-for-Buck equalization (MRS equals the price ratio); and
- utility is just equal to $\bar{u}$ (the constraint binds).

Theorem 4.3.2 (Properties of the Expenditure Function). *If u is continuous and strictly increasing, then the expenditure function $e(\mathbf{p}, \bar{u})$ is:*

1. 0 when $\bar{u} = u(0)$,
2. continuous on its domain,
3. strictly increasing and unbounded above in the utility level $\bar{u}$,
4. increasing in $\mathbf{p}$,
5. homogeneous of degree 1 in $\mathbf{p}$,
6. concave in $\mathbf{p}$.

Theorem 4.3.3 (Duality Relations). *If u is continuous and strictly increasing, then*

$$e(\mathbf{p}, v(\mathbf{p}, I)) = I, \quad v(\mathbf{p}, e(\mathbf{p}, \bar{u})) = \bar{u}.$$

Theorem 4.3.4 (Marshallian–Hicksian Link). *If u is strictly increasing, strictly quasiconcave, and continuous, then*

$$\mathbf{x}(\mathbf{p}, I) = \mathbf{x}^h(\mathbf{p}, v(\mathbf{p}, I)), \quad \mathbf{x}^h(\mathbf{p}, \bar{u}) = \mathbf{x}(\mathbf{p}, e(\mathbf{p}, \bar{u})).$$

4.4 Consumer Welfare

Recall from first year micro.

Consumer surplus = area under demand – total expenditure.

Welfare change is often proxied by ΔCS .

Remark. This is exact only under special conditions (notably quasi-linear utility / negligible income effects).

General welfare analysis asks a money-metric question: How much money are you willing to pay to move from situation A to situation B?

This can be triggered by:

- Price changes,
- Income changes,
- Any policy/environment change affecting utility.

Example 4.4.1. Take utility

$$u(x_1, x_2) = \ln x_1 + x_2,$$

with budget

$$p_1 x_1 + p_2 x_2 \leq I, \quad x_1, x_2 \geq 0.$$

We study welfare effects of a change in p_1 .

Step 1 (FOCs and corner logic)

Set up the Lagrangian:

$$\mathcal{L} = \ln x_1 + x_2 + \lambda (I - p_1 x_1 - p_2 x_2).$$

For an interior solution with $x_2 > 0$, the FOCs are

$$\frac{1}{x_1} = \lambda p_1, \quad 1 = \lambda p_2 \Rightarrow x_1 = \frac{p_2}{p_1}.$$

Then

$$x_2 = \frac{I - p_1 x_1}{p_2} = \frac{I}{p_2} - 1,$$

which requires $I \geq p_2$ for feasibility ($x_2 \geq 0$).

Step 2 (Piecewise form): full Marshallian demand

Combining the interior-case solution with the corner case, the full Marshallian demand is:

$$x_1(p_1, p_2, I) = \begin{cases} \frac{I}{p_1}, & I < p_2, \\ \frac{p_2}{p_1}, & I \geq p_2, \end{cases} \quad x_2(p_1, p_2, I) = \begin{cases} 0, & I < p_2, \\ \frac{I}{p_2} - 1, & I \geq p_2. \end{cases}$$

For $I \geq p_2$, demand for good 1 is income-independent. This is the no-income-effect feature used in the welfare result.

Fix I and p_2 . In the high-income region $I \geq p_2$,

$$x_1(p_1) = \frac{p_2}{p_1} \quad (\text{equivalently } p_1 = \frac{p_2}{x_1}).$$

Fix $I = 80$ and $p_2 = 40$. The price of good 1 changes from $p_1 = 40$ to $p'_1 = 20$.

Since $I \geq p_2$, we are in the high-income region where

$$x_1(p_1) = \frac{p_2}{p_1}.$$

Thus

$$x_1 = \frac{40}{40} = 1, \quad x'_1 = \frac{40}{20} = 2.$$

$$\Delta CS = \int_{p'_1}^{p_1} x_1(p_1) dp_1 = \int_{20}^{40} \frac{40}{p_1} dp_1 = 40 \ln 2 \approx 27.73.$$

Rectangle + triangle approximation gives about 30 (slight overestimate).

Initial situation $(p_1, p_2, I) = (40, 40, 80)$:

$$(x_1, x_2) = (1, 1), \quad u = \ln 1 + 1 = 1.$$

After the price drop to $p'_1 = 20$:

$$(x'_1, x'_2) = (2, 1).$$

Now ask: how much income m can be taken away at new prices while keeping utility at 1?

At new prices $(20, 40)$, with income reduced to $80 - m$:

$$x'_1 = 2 \quad (\text{unchanged; no income effect for good 1}).$$

$$x'_2 = \frac{80 - m}{40} - 1 = 1 - \frac{m}{40}.$$

So

$$u' = \ln 2 + 1 - \frac{m}{40}.$$

Equalize to old utility $u = 1$:

$$\ln 2 + 1 - \frac{m}{40} = 1 \Rightarrow m = 40 \ln 2.$$

Hence willingness to pay equals $40 \ln 2$, exactly matching ΔCS .

In this quasi-linear setup, good 1 has no income effect in the relevant region. So the area under demand captures the exact money-metric welfare change. Result in this example:

$$\Delta CS = CV = 40 \ln 2.$$

Definition 4.4.2 (Compensating Variation (CV)). For a change from $(\mathbf{p}, I)$ to $(\mathbf{p}', I')$, with

$$u = u(\mathbf{x}(\mathbf{p}, I)), \quad u' = u(\mathbf{x}(\mathbf{p}', I')),$$

define

$$CV = e(\mathbf{p}', u') - e(\mathbf{p}', u) = I' - e(\mathbf{p}', u).$$

It measures how much income can be taken away at new prices while leaving the person at old utility.

To interpret compensating variation (CV):

- $CV > 0$: the new situation is better (the consumer is willing to pay for it).
- $CV < 0$: the new situation is worse (the consumer needs compensation).

A sign-convention note is important here: JR defines CV with the opposite sign. These slides use the more common convention in which positive CV means there has been a welfare gain.

Remark. CV is the correct money-metric welfare measure. By contrast, ΔCS is exact only when income effects are negligible, as in the quasi-linear benchmark. When income effects are substantial, ΔCS can either overstate or understate the true welfare change.

Why ΔCS can be biased: The reason is that ΔCS is obtained by integrating Marshallian demand. Marshallian demand contains both substitution effects and income effects. Therefore, if marginal willingness to pay changes strongly with wealth, the area-under-demand measure becomes biased as a welfare measure.

Definition 4.4.3 (Equivalent Variation). An alternative money-metric welfare measure is Equivalent Variation (EV), defined as

$$EV = e(\mathbf{p}, u') - e(\mathbf{p}, u).$$

This uses old prices $\mathbf{p}$ and new utility u' .

This contrasts with Compensating Variation (CV):

- EV uses old prices $\mathbf{p}$ and new utility u' .
- CV uses new prices $\mathbf{p}'$ and old utility u .

Both EV and CV are theoretically valid money-metric welfare measures.

Putting the main ideas together:

- Demand is naturally a correspondence; uniqueness requires curvature, specifically strict quasi-concavity.
- Demand has a robust structure: it is non-empty, satisfies budget exhaustion, and is homogeneous of degree 0 in $(\mathbf{p}, I)$.
- Indirect utility is continuous, increasing in income, decreasing in prices, and quasi-convex.
- The utility-maximization problem (U-max) and the expenditure-minimization problem (E-min) are dual to each other; this duality relies on continuity and monotonicity assumptions.
- In welfare analysis, CV and EV are money-metric measures, while consumer surplus (CS) is exact in the quasi-linear benchmark and only approximate otherwise.

5 Uncertainty and Expected Utility

Uncertainty appears in many familiar situations.

Example 5.0.1. The contents of a box of chocolate are uncertain, the realization of exam questions is uncertain, an opponent's behavior in a game is uncertain, the financial payoff from risky assets is uncertain, and the outcomes of lottery tickets are uncertain.

These examples show that many economic decisions are made under uncertainty rather than under certainty.

In previous weeks, the alternatives were bundles $\mathbf{x} \in \mathbb{R}_+^n$. We studied preferences over bundles, denoted by $\succeq$, and then obtained a utility representation. The consumer's problem was to maximize utility over the budget set. However, we now move from certain outcomes to uncertain outcomes. Because of this shift, we need a new set of alternatives, we need axioms that are appropriate for that set, and we need a representation that is consistent with those axioms.

This leads to three guiding questions:

1. What is the right set of alternatives under uncertainty?
2. What axioms should preferences satisfy on that set?
3. What representation form should utility take?

Under uncertainty, the primitive alternatives are not certain bundles anymore. Instead, let

$$Z = \{z_1, \dots, z_n\}$$

be the set of outcomes. In this course, Z is assumed to be finite, although the results can be extended to continuous probability distributions. The elements of Z may be bundles of goods, amounts of money, or complete descriptions of a situation.

Therefore, the relevant set of alternatives is the set of lotteries L over outcomes in Z , and individual lotteries are denoted by lowercase letters such as $\mathbf{p}$ and $\mathbf{q}$.

Definition 5.0.2. A lottery is a probability distribution over Z . Formally,

$$\mathbf{p} = (p_1, \dots, p_n), \quad p_i \geq 0, \quad \sum_{i=1}^n p_i = 1.$$

Accordingly, the set of all lotteries is

$$L = \Delta(Z) = \{\mathbf{p} \in \mathbb{R}_+^n : \sum_i p_i = 1\}.$$

This means that preferences under uncertainty are defined over lotteries, that is, $\succeq$ is defined on L .

When we want to describe a lottery over only part of the outcome set, we can write it in the form

$$(z_1, p_1; \dots; z_n, p_n).$$

Example 5.0.3. $(z_1, 0.2; z_2, 0.8)$ means that z_1 occurs with probability 20% and z_2 occurs with probability 80%. This notation makes the probabilistic structure of the alternative explicit.

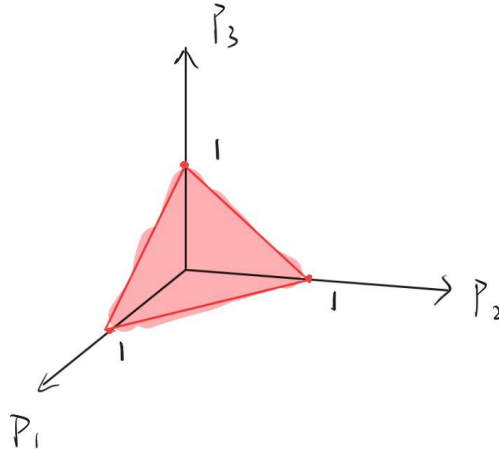


Figure 1: The geometry of the lottery set.

When $n = 3$, the set L is a triangle, or equivalently a 2-simplex. This set is compact, convex, and a subset of $\mathbb{R}_+^3$. In the diagram, the three vertices are labeled $(1, 0, 0)$, $(0, 1, 0)$, and $(0, 0, 1)$, and the point $\mathbf{p}$ represents one particular lottery in the set.

At this point, there are two different frameworks for decision under uncertainty.

1. In the expected utility framework studied in this course, the primitive objects are lotteries over outcomes, so a lottery is written as

$$\mathbf{p} : Z \rightarrow [0, 1].$$

2. By contrast, in subjective expected utility, which is outside the scope of the course here, the primitive objects are acts that map states to outcomes, written as

$$f : S \rightarrow Z.$$

This contrast matters because the two theories use different primitive objects and therefore require different axiomatizations. At the same time, they share a similar representation logic. In the subjective expected utility framework, preferences over acts also reveal the decision maker's subjective beliefs. Thus, the lecture is clarifying that today's analysis is about lotteries over outcomes, not acts over states.

5.1 St. Petersburg and Expected Value

The St. Petersburg setup asks the decision maker to choose between two alternatives. Option A is to receive \$1000 for sure. Option B is a gamble in which a coin is tossed until the first tail appears on toss n , and the payoff is 2^n . What criterion should describe choice?

Definition 5.1.1. For a lottery $\mathbf{p}$, expected value is defined as

$$EV(\mathbf{p}) = \sum_{z \in Z} p_z z.$$

The associated rule is simple: choose the lottery with the higher expected value. This gives a direct way to rank lotteries when outcomes can be evaluated numerically.

Example 5.1.2. Suppose lottery $\mathbf{p}$ gives a gain of 100 with probability 1/2 and a loss of 50 with probability 1/2. Then

$$EV(\mathbf{p}) = 0.5(100) + 0.5(-50) = 25.$$

Now compare this with lottery $\mathbf{q}$, which gives 40 for sure. In that case,

$$EV(\mathbf{q}) = 40.$$

Since $40 > 25$, the expected-value criterion ranks $\mathbf{q}$ above $\mathbf{p}$.

In the St. Petersburg game, to receive payoff 2^n , the coin must land heads on the first $n - 1$ tosses and tails on the n -th toss. Because the flips are independent, the probability of this event is

$$2^{-n}.$$

Therefore, the expected value of option B is

$$EV(B) = \sum_{n=1}^{\infty} 2^{-n} \cdot 2^n = \sum_{n=1}^{\infty} 1 = \infty.$$

So, under the expected-value criterion, option B has infinite expected value.

According to expected value, one should choose option B over any finite sure amount. However, behaviorally, most people prefer some finite sure amount instead. This mismatch between the prediction of expected value and observed choice motivates replacing expected value with expected utility.

Bernoulli's resolution begins by identifying what the expected-value criterion assumes about utility. If choices are based only on expected monetary value, then utility must effectively be linear in wealth, that is,

$$u(w) = w.$$

Bernoulli argued instead that the marginal value of wealth diminishes. He expressed this by

$$\frac{du}{dw} = \frac{1}{w} \quad \Rightarrow \quad u(w) = \ln w.$$

Therefore, lotteries should be evaluated by $E[\ln w]$ rather than by $E[w]$.

Applying this idea to the St. Petersburg game changes the ranking of the two options. For option A , which gives \$1000 for sure,

$$EU(A) = \ln(1000).$$

For option B , the payoff is 2^n when the first tail appears on toss n , which occurs with probability 2^{-n} . Hence

$$EU(B) = \sum_{n=1}^{\infty} 2^{-n} \ln(2^n) = \sum_{n=1}^{\infty} \frac{n \ln 2}{2^n} = 2 \ln 2 = \ln 4.$$

This calculation uses the fact that

$$\sum_{n=1}^{\infty} \frac{n}{2^n} = 2.$$

Since

$$\ln(1000) > \ln 4 \quad \Rightarrow \quad EU(A) > EU(B).$$

So under logarithmic utility, the sure \$1000 is preferred to the St. Petersburg gamble.

The paradox disappears once utility is concave. Concavity captures diminishing marginal utility of wealth, and this makes expected utility a more plausible criterion than expected value for describing actual behavior under risk.

5.2 Expected Utility Criterion

Now we generalize the idea into the full expected utility framework.

Definition 5.2.1. Let $Z = \{z_1, \dots, z_n\}$ be the set of outcomes. A person is an expected utility maximizer if there exists a utility function $u : Z \rightarrow \mathbb{R}$ such that

$$\mathbf{p} \succeq \mathbf{q} \iff \sum_{i=1}^n p_i u(z_i) \geq \sum_{i=1}^n q_i u(z_i).$$

We say that u represents the preference relation $\succeq$ over the lottery set L .

This means that lotteries are ranked by the expected value of utility across outcomes, rather than by expected monetary value alone.

Rewrites this representation more formally by defining

$$V(\mathbf{p}) = \sum_{i=1}^n p_i u(z_i).$$

With this notation,

$$\mathbf{p} \succeq \mathbf{q} \iff V(\mathbf{p}) \geq V(\mathbf{q}).$$

This makes clear that the object directly representing preferences over lotteries is V , since V is defined on lotteries, while u is defined only on certain outcomes. Still, under expected utility, V is linear in probabilities, so knowing u is equivalent to knowing V . This is why the lecture can sometimes speak loosely of u representing preferences over lotteries.

Example 5.2.2 (Simple Numerical Example). The outcomes are

$$\mathbf{z}_1 = (1, 1), \quad \mathbf{z}_2 = (1, 2), \quad \mathbf{z}_3 = (3, 0),$$

so that

$$Z = \{\mathbf{z}_1, \mathbf{z}_2, \mathbf{z}_3\}.$$

Take the utility function

$$u(x_1, x_2) = x_1 x_2.$$

Then the utility of each outcome is

$$u(\mathbf{z}_1) = 1, \quad u(\mathbf{z}_2) = 2, \quad u(\mathbf{z}_3) = 0.$$

Now consider the lottery

$$\mathbf{p} = \left(\frac{1}{2}, \frac{1}{3}, \frac{1}{6}\right).$$

Its expected utility is

$$EU(\mathbf{p}) = \frac{1}{2} \cdot 1 + \frac{1}{3} \cdot 2 + \frac{1}{6} \cdot 0 = \frac{7}{6}.$$

So the example shows exactly how expected utility is computed: first evaluate each outcome with u , then take the probability-weighted average.

Receiving $\mathbf{z}_1$ for sure corresponds to the lottery $(1, 0, \dots, 0)$. Thus, certainty is embedded in the same object class as risk, rather than being treated separately. More generally, any certain outcome $\mathbf{z}_i$ can be represented by the degenerate lottery that assigns probability 1 to $\mathbf{z}_i$ and probability 0 to all other outcomes.

Definition 5.2.3. A compound lottery is a lottery over lotteries.

This means that uncertainty can arise in multiple stages, since one first faces uncertainty over which lottery is selected and then uncertainty within the selected lottery itself. The expected utility framework handles this by reducing compound lotteries to simple lotteries.

Example 5.2.4. With probability $1/2$, one draws lottery $\mathbf{p}$, and with probability $1/2$, one draws lottery $\mathbf{q}$.

The reduction principle says that a compound lottery such as

$$\frac{1}{2}\mathbf{p} + \frac{1}{2}\mathbf{q}$$

can be represented as a simple lottery whose final outcome probabilities are weighted averages of the probabilities in $\mathbf{p}$ and $\mathbf{q}$. The lecture's example is

$$\frac{1}{2} \left(\frac{1}{2}, \frac{1}{3}, \frac{1}{6} \right) + \frac{1}{2} (1, 0, 0) = \left(\frac{3}{4}, \frac{1}{6}, \frac{1}{12} \right).$$

Coordinate by coordinate, this works as follows:

$$\frac{1}{2} \left(\frac{1}{2}, \frac{1}{3}, \frac{1}{6} \right) = \left(\frac{1}{4}, \frac{1}{6}, \frac{1}{12} \right), \quad \frac{1}{2} (1, 0, 0) = \left(\frac{1}{2}, 0, 0 \right),$$

and adding them gives

$$\left(\frac{1}{4}, \frac{1}{6}, \frac{1}{12} \right) + \left(\frac{1}{2}, 0, 0 \right) = \left(\frac{3}{4}, \frac{1}{6}, \frac{1}{12} \right).$$

So the reduced simple lottery assigns probability 3/4 to the first outcome, 1/6 to the second, and 1/12 to the third.

5.3 Axioms for EU Representation

The central question is: why can we not simply reuse all the axioms from the certainty model? The answer is that the environment has changed. The alternatives are now lotteries, not bundles, so some axioms transfer naturally while others do not.

L , the set of lotteries, is still a large subset of $\mathbb{R}_+^n$,

- (i) so continuity remains a natural requirement.
- (ii) However, strong monotonicity has little bite in the simplex, because probabilities must sum to 1, so increasing one probability necessarily decreases another.
- (iii) In addition, convexity restrictions can be too strong in this context. The reason is that the alternatives are lotteries rather than outcomes, and a convex combination of lotteries is simply a compound lottery.

Therefore, the practical route is not to reuse all certainty axioms mechanically, but instead to start from a choice criterion—expected utility—and derive the axioms that identify it.

Because $L \subset \mathbb{R}_+^n$ is closed and convex, if preferences on L are complete, transitive, and continuous, then they admit a continuous utility representation on L .

However, expected utility requires more than mere continuity. Specifically, it requires linearity in probabilities. So a general continuous utility representation is not yet enough to obtain the expected utility form.

Axiom 5.3.1 (A1: Completeness and Transitivity). Completeness means that any two lotteries are comparable. Transitivity means that rankings are internally consistent.

These are baseline rationality axioms, playing the same role here as they do in the certainty model.

Axiom 5.3.2 (A2: Continuity). For each lottery $\mathbf{p} \in L$, the upper and lower contour sets are closed:

$$G(\mathbf{p}) = \{\mathbf{q} \in L : \mathbf{q} \succeq \mathbf{p}\}, \quad W(\mathbf{p}) = \{\mathbf{q} \in L : \mathbf{p} \succeq \mathbf{q}\}.$$

This axiom rules out abrupt jumps in preferences. An equivalent intuition: if z_1 is the worst certain outcome and z_n is the best certain outcome, then for any lottery $\mathbf{l}$, there exists some probability $p \in [0, 1]$ such that

$$\mathbf{l} \sim (z_1, p; z_n, 1 - p).$$

So every lottery can be matched, in terms of preference, by an appropriate mixture of the worst and best certain outcomes.

Axiom 5.3.3 (A3: Independence). For all $\mathbf{p}, \mathbf{q}, \mathbf{r} \in L$ and all $\alpha \in (0, 1)$,

$$\mathbf{p} \succeq \mathbf{q} \iff \alpha \mathbf{p} + (1 - \alpha) \mathbf{r} \succeq \alpha \mathbf{q} + (1 - \alpha) \mathbf{r}.$$

This means that mixing both alternatives with the same third lottery preserves their ranking. If $\mathbf{p}$ is preferred to $\mathbf{q}$, then adding the same background risk $\mathbf{r}$ in the same proportion should not reverse that preference. The lecture notes that JR splits this axiom into two parts, substitution and reduction.

Theorem 5.3.4 (Expected Utility Theorem). *preferences over lotteries satisfy A1, A2, and A3 if and only if they admit an expected utility representation. Formally,*

$$\mathbf{p} \succeq \mathbf{q} \iff \sum_i p_i u(z_i) \geq \sum_i q_i u(z_i).$$

If two utility functions u and v represent the same expected utility preferences, then they must satisfy

$$u = \alpha v + \beta, \quad \alpha > 0.$$

So an expected utility representation is unique only up to a positive affine transformation.

This is stronger than the certainty case, because ordinary ordinal utility over bundles is preserved under any strictly increasing transformation, whereas expected utility allows only affine changes.

The implication is that expected utility imposes much more structure than a generic ordinal representation.

5.4 Risk Attitudes

Let $Z \subseteq \mathbb{R}$, so outcomes are money levels.

Definition 5.4.1. A person is

1. strictly risk averse if they prefer the sure amount equal to the lottery's expected value to any risky lottery with the same mean;
2. risk neutral if they are indifferent between the risky lottery and the sure amount with the

same mean;

3. strictly risk loving if they prefer the risky lottery to the sure amount with the same mean.

For an expected utility maximizer,

- (i) strictly concave $u \iff$ strict risk aversion
- (ii) linear $u \iff$ risk neutrality
- (iii) and strictly convex $u \iff$ strict risk loving.

Thus, risk attitude is fully encoded in the curvature of utility.

For a concave utility function, the utility of the mean wealth is at least as large as the mean of utility, and for a strictly concave utility function with a non-degenerate lottery, the inequality is strict:

$$u(E[w]) \geq E[u(w)],$$

with

$$u(E[w]) > E[u(w)]$$

under strict concavity and non-degeneracy. This is the Jensen intuition. It explains why a risk-averse person prefers the sure amount equal to the lottery's expected value rather than the risky lottery itself.

Definition 5.4.2. For a lottery $\mathbf{p}$, the certainty equivalent CE is defined by

$$u(CE) = EU(\mathbf{p}),$$

so CE is the sure amount that makes the person indifferent to the lottery. The risk premium is

$$\pi = EV(\mathbf{p}) - CE.$$

Under strict risk aversion,

$$\pi > 0.$$

Equivalently, a strictly risk-averse person has

$$CE < EV(\mathbf{p}),$$

meaning they would accept less than the expected value in exchange for certainty.

Example 5.4.3. Consider a lottery that yields $w_H = 100$ and $w_L = 0$, each with probability $1/2$. Its expected value is

$$EV = \frac{1}{2} \cdot 100 + \frac{1}{2} \cdot 0 = 50.$$

Now suppose utility is given by

$$u(w) = \sqrt{w}.$$

Then expected utility is

$$EU = \frac{1}{2}\sqrt{100} + \frac{1}{2}\sqrt{0} = 5.$$

The certainty equivalent CE is the sure amount that gives the same utility as the lottery, so it is

defined by

$$u(CE) = EU.$$

Since $u(w) = \sqrt{w}$, this means

$$\sqrt{CE} = 5,$$

and therefore

$$CE = 25.$$

The risk premium is then

$$\pi = EV - CE = 50 - 25 = 25.$$

This example shows concretely that with a concave utility function, the certainty equivalent lies below expected value, so the risk premium is positive.

5.5 Summary

Under uncertainty, the primitive alternatives are lotteries over outcomes. The expected-value criterion is not sufficient, because it fails in the St. Petersburg case. The expected utility criterion instead ranks lotteries by the expected utility of outcomes. The axioms A1, A2, and A3 characterize when such an expected utility representation exists. That representation is unique only up to positive affine transformations. Finally, when outcomes are money levels, the curvature of utility maps directly into risk attitude.

Putting these pieces together, the lecture's main message is that choice under uncertainty requires a different framework from choice under certainty. The correct objects are lotteries, not bundles. Expected value is too crude because it ignores diminishing marginal utility. Expected utility provides the right representation, its axioms explain exactly when it applies, and its curvature properties explain why different people may be risk averse, risk neutral, or risk loving. The two-outcome example at the end confirms all of this numerically in one place.

6 Producer Theory

Producer theory mirrors consumer theory in many ways, but not completely. The main differences are important.

- First, production/output has cardinal meaning, so output levels can be directly interpreted and compared quantitatively. By contrast, utility does not usually have cardinal meaning in the same way.
- Second, the firm does not usually face an income constraint in the same way as the consumer.

These differences make producer theory similar in structure to consumer theory, while still requiring its own concepts and tools.

6.1 Producer Setup

A firm acquires inputs and combines them to produce output. Inputs are purchased on input markets, and these expenditures are the firm's costs. Outputs are sold on product markets, and these sales generate revenue. The standard benchmark assumption is that firms maximize profit.

Definition 6.1.1 (Production possibility set and production plan). To describe production more generally, a firm can be represented by a production possibility set $Y \subseteq \mathbb{R}^m$. Each vector $\mathbf{y} = (y_1, \dots, y_m) \in Y$ is called a production plan. The convention is as follows:

- if $y_i < 0$, resource i is an input;
- if $y_i > 0$, resource i is an output.

This language is especially useful when there are many inputs and outputs, because it provides a single general framework for describing all feasible production plans.

After introducing the general production possibility set, we can focus on the case of a single output. For one output y and an input vector $\mathbf{x} = (x_1, \dots, x_n) \in \mathbb{R}_+^n$, a production function is written as

$$y = f(\mathbf{x}).$$

The interpretation is that $f(\mathbf{x})$ gives the maximum output producible from the input vector $\mathbf{x}$. In other words, the production function summarizes the firm's technology by showing the greatest feasible output associated with each input bundle.

Definition 6.1.2 (Assumption 3.1). The production function $f : \mathbb{R}_+^n \rightarrow \mathbb{R}_+$ is assumed to be continuous, strictly increasing, and strictly quasiconcave, and it satisfies $f(\mathbf{0}) = 0$.

Example 6.1.3 (Examples with Two Inputs). To make the definition concrete, consider three standard examples with two inputs x_1 and x_2 .

For perfect complements,

$$f(x_1, x_2) = \min\{x_1, x_2\}.$$

For perfect substitutes,

$$f(x_1, x_2) = x_1 + x_2.$$

For the Cobb–Douglas production function,

$$f(x_1, x_2) = x_1^\alpha x_2^\beta,$$

usually with $\alpha + \beta < 1$.

Once the production function is defined, the next step is to study how output changes when all inputs are scaled up by the same factor. This is the idea of returns to scale.

Definition 6.1.4 (Returns to Scale). • If

$$f(t\mathbf{x}) = tf(\mathbf{x}) \quad \forall t > 0,$$

the technology has constant returns to scale.

- If

$$f(t\mathbf{x}) < tf(\mathbf{x}) \quad \forall t > 1,$$

the technology has decreasing returns to scale.

- If

$$f(t\mathbf{x}) > tf(\mathbf{x}) \quad \forall t > 1,$$

the technology has increasing returns to scale.

Question. Why is Cobb–Douglas usually restricted by $\alpha + \beta < 1$ or $\alpha + \beta \leq 1$?

Answer. For the Cobb–Douglas function, $f(x_1, x_2) = x_1^\alpha x_2^\beta$, if all inputs are multiplied by t , then

$$f(tx_1, tx_2) = (tx_1)^\alpha (tx_2)^\beta = t^{\alpha+\beta} f(x_1, x_2).$$

Therefore:

- if $\alpha + \beta = 1$, the function has constant returns to scale;
- if $\alpha + \beta < 1$, it has decreasing returns to scale;
- if $\alpha + \beta > 1$, it has increasing returns to scale.

Economists often restrict the model to $\alpha + \beta \leq 1$ because, with increasing returns to scale, competitive profit maximization can behave badly. In particular, average cost may keep falling, and a finite profit-maximizing output may fail to exist. The statement $\alpha + \beta < 1$ is simply a stronger restriction than $\alpha + \beta \leq 1$; the broader economic reason is to avoid the problems created by increasing returns.

Definition 6.1.5 (Marginal Product). When the production function f is differentiable, the partial derivative

$$\frac{\partial f(\mathbf{x})}{\partial x_i}$$

is called the marginal product of input i . It measures how much output increases when input i rises by a small amount, holding the other inputs fixed.

A standard assumption is that marginal products are strictly positive, which means that more input raises output. This is consistent with the earlier assumption that the production function is strictly increasing.

Definition 6.1.6 (Isoquants). For a fixed output level y , the y -level isoquant is

$$Q(y) = \{\mathbf{x} \geq \mathbf{0} : f(\mathbf{x}) = y\}.$$

An isoquant is therefore a level set of the production function.

Every point on the isoquant represents a different input combination that produces exactly the same output level y . In the two-input case, the isoquant is drawn in (x_1, x_2) -space, and moving along it shows how the firm can trade off one input against the other while holding output fixed.

Definition 6.1.7 (Marginal Rate of Technical Substitution). The marginal rate of technical substitution of input j for input i is

$$MRTS_{ij}(\mathbf{x}) = \frac{\partial f(\mathbf{x})/\partial x_i}{\partial f(\mathbf{x})/\partial x_j}.$$

It measures how one input can replace another while holding output fixed.

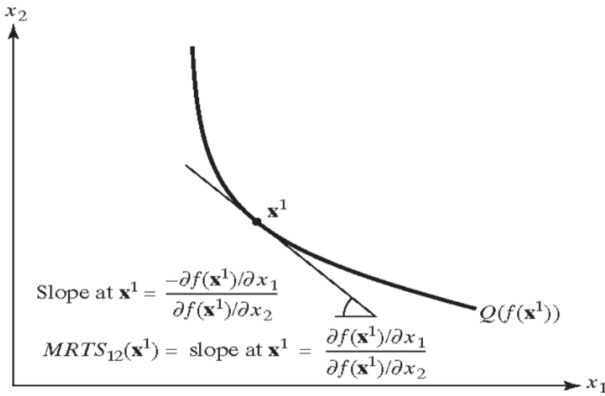


Figure 2: The marginal rate of technical substitution.

In the two-input case, the MRTS is the absolute value of the slope of the isoquant.

Proof. For a fixed output level y , the isoquant satisfies

$$f(x_1, x_2) = y.$$

Moving along this isoquant, output does not change, so the total differential is equal to 0:

$$df = \frac{\partial f}{\partial x_1} dx_1 + \frac{\partial f}{\partial x_2} dx_2 = 0.$$

Rearranging gives

$$\frac{dx_2}{dx_1} = -\frac{\partial f/\partial x_1}{\partial f/\partial x_2}.$$

By definition,

$$MRTS_{12}(\mathbf{x}) = \frac{\partial f/\partial x_1}{\partial f/\partial x_2}.$$

Therefore, we immediately obtain

$$MRTS_{12}(\mathbf{x}) = -\frac{dx_2}{dx_1}.$$

□

6.2 Cost Minimization

Theorem 6.2.1. *The production set can be written as*

$$Y = \{(-\mathbf{x}, y) : y \leq f(\mathbf{x}), \mathbf{x} \in \mathbb{R}_+^n\}.$$

It is convex if and only if the production function $f(\mathbf{x})$ is concave.

This result links the geometric property of the production set to the curvature property of the production function. Convex production sets are convenient for equilibrium theory, although they are not always empirically reasonable.

Assume that firms are perfectly competitive in input markets. The input price vector is

$$\mathbf{w} = (w_1, \dots, w_n) > 0.$$

The firm takes these prices as given. For any input vector $\mathbf{x}$, the cost of using that bundle is

$$\mathbf{w} \cdot \mathbf{x} = \sum_{i=1}^n w_i x_i.$$

Thus, total input cost is the sum of the expenditures on each individual input.

With output price p , profit can be written as

$$\pi = py - c(\mathbf{w}, y).$$

Here, $c(\mathbf{w}, y)$ denotes the cost of producing output y when the input price vector is $\mathbf{w}$. For any chosen output level y , the firm wants the cheapest input bundle that can produce that output. Otherwise, it could reduce cost and increase profit while holding output fixed.

Definition 6.2.2. For all strictly positive input price vectors $\mathbf{w} > 0$ and output levels $y \in f(\mathbb{R}_+^n)$, the cost function is defined as

$$c(\mathbf{w}, y) = \min_{\mathbf{x} \in \mathbb{R}_+^n} \mathbf{w} \cdot \mathbf{x} \quad \text{s.t.} \quad f(\mathbf{x}) \geq y.$$

This means that $c(\mathbf{w}, y)$ is the minimum cost of producing at least output level y when the firm faces input prices $\mathbf{w}$. If $\mathbf{x}(\mathbf{w}, y)$ solves this problem, then

$$c(\mathbf{w}, y) = \mathbf{w} \cdot \mathbf{x}(\mathbf{w}, y).$$

Because the production function f is strictly increasing, the output constraint must bind at the optimum. Therefore, the problem

$$\min_{\mathbf{x} \in \mathbb{R}_+^n} \mathbf{w} \cdot \mathbf{x} \quad \text{s.t.} \quad f(\mathbf{x}) \geq y$$

is equivalent to

$$\min_{\mathbf{x} \in \mathbb{R}_+^n} \mathbf{w} \cdot \mathbf{x} \quad \text{s.t.} \quad f(\mathbf{x}) = y.$$

The reason is straightforward: producing more than the target output would waste cost.

If f is strictly quasiconcave, then the cost-minimizing input vector is unique.

Definition 6.2.3. This vector is denoted by

$$\mathbf{x}(\mathbf{w}, y),$$

and it is called the firm's conditional input demand.

The term “conditional” is important: it means that the input demand is derived conditional on the output level y . At this stage, the output level is taken as given, and it need not yet be the profit-maximizing output level.

Assume an interior solution and differentiability. The Lagrangian is

$$\mathcal{L} = \mathbf{w} \cdot \mathbf{x} + \lambda(y - f(\mathbf{x})).$$

The first-order conditions are

$$w_i = \lambda \frac{\partial f(\mathbf{x}^*)}{\partial x_i}, \quad i = 1, \dots, n.$$

Thus, for any two inputs i and j ,

$$\frac{\partial f(\mathbf{x}^*)/\partial x_i}{\partial f(\mathbf{x}^*)/\partial x_j} = \frac{w_i}{w_j}.$$

This means that, at the cost-minimizing input bundle, the ratio of marginal products must equal the ratio of input prices.

Cost minimization implies

$$MRTS_{ij}(\mathbf{x}) = \frac{\partial f(\mathbf{x})/\partial x_i}{\partial f(\mathbf{x})/\partial x_j} = \frac{w_i}{w_j}.$$

The economic meaning is that the technical tradeoff between inputs must match their market-price tradeoff. If this equality did not hold, the firm could substitute away from the relatively expensive input toward the relatively cheap one and reduce cost while keeping output fixed. This is the producer-side analogue of the tangency condition in consumer theory.

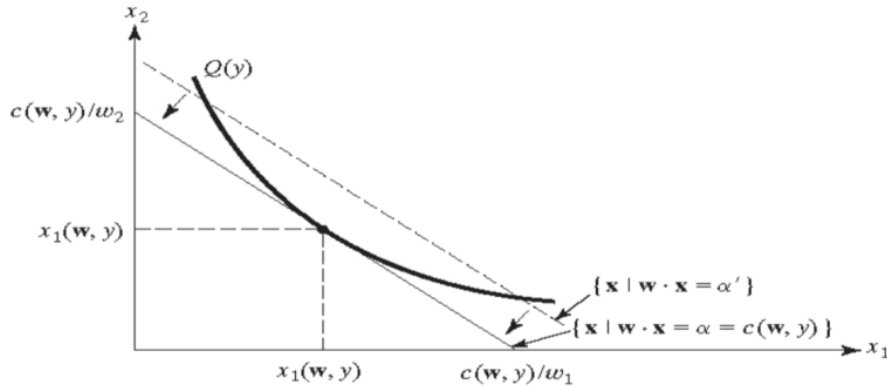


Figure 3: The solution to the firms' cost-minimisation problem.

6.2.1 Examples of Cost Functions

Example 6.2.4 (Perfect Complements Cost Function). For the perfect-complements technology

$$f(x_1, x_2) = \min\{x_1, x_2\},$$

the firm must use the two inputs in fixed proportion. To produce output y , it needs

$$x_1 = x_2 = y.$$

Any bundle with one input below y cannot produce y , and any bundle with one input above y wastes cost without raising output. Therefore, the cost function is

$$c(\mathbf{w}, y) = (w_1 + w_2)y.$$

When solving examples, the final answer should be written as a function of output y and prices, not in terms of the chosen input quantities. In other words, the final object must be the cost function $c(\mathbf{w}, y)$, not just the optimal bundle (x_1, x_2) .

Example 6.2.5 (Perfect Substitutes Cost Function). For the perfect-substitutes technology

$$f(x_1, x_2) = x_1 + x_2,$$

the two inputs can replace each other one for one. To produce output y , the firm only needs a bundle satisfying

$$x_1 + x_2 = y.$$

Since the inputs are perfect substitutes, the firm uses only the cheaper input. Hence,

$$c(\mathbf{w}, y) = \min\{w_1, w_2\}y.$$

Example 6.2.6 (Cobb–Douglas Cost Function). For the Cobb–Douglas production function

$$f(x_1, x_2) = x_1^\alpha x_2^\beta,$$

the cost function takes the form

$$c(\mathbf{w}, y) = A(\alpha, \beta) w_1^{\frac{\alpha}{\alpha+\beta}} w_2^{\frac{\beta}{\alpha+\beta}} y^{\frac{1}{\alpha+\beta}},$$

where

$$A(\alpha, \beta) = \left(\frac{\alpha}{\beta}\right)^{\frac{\beta}{\alpha+\beta}} + \left(\frac{\beta}{\alpha}\right)^{\frac{\alpha}{\alpha+\beta}} = (\alpha + \beta) \alpha^{-\frac{\alpha}{\alpha+\beta}} \beta^{-\frac{\beta}{\alpha+\beta}}.$$

Once input prices $\mathbf{w}$ and the technological parameters α and β are fixed, the cost function can be written in the simplified form

$$c(\mathbf{w}, y) = Ay^r, \quad r = \frac{1}{\alpha + \beta}, \quad A := A(\alpha, \beta) w_1^{\frac{\alpha}{\alpha+\beta}} w_2^{\frac{\beta}{\alpha+\beta}},$$

where A is a constant with respect to output y . This form is useful because it makes the curvature of the cost function depend entirely on the exponent r .

If $\alpha + \beta = 1$, then $r = 1$, so $c(\mathbf{w}, y) = Ay$. In this case, cost is linear in output, which means marginal cost is constant.

If $\alpha + \beta = \frac{1}{2}$, then $r = 2$, so $c(\mathbf{w}, y) = Ay^2$. In this case, marginal cost rises linearly with output.

More generally, the curvature of the cost function depends on the value of r . If $r > 1$, cost is convex in output and marginal cost rises with output. If $r = 1$, cost is linear and marginal cost is constant. If $r < 1$, cost is concave in output and marginal cost falls with output.

Why $r < 1$ causes trouble? Since $r = \frac{1}{\alpha+\beta}$, the condition $\alpha+\beta > 1$ implies $r < 1$. This corresponds to increasing returns to scale in production. The cost function is concave in output, so marginal cost falls as output expands.

This creates problems for competitive profit maximization. If profit is written as

$$\pi(y) = py - Ay^r,$$

then with $0 < r < 1$, the cost term grows more slowly than the revenue term. As a result, firms have an incentive to expand output too aggressively, and a finite interior profit maximum may fail to exist.

6.2.2 Properties of Cost Function

Theorem 6.2.7 (Properties of the Cost Function). *If f is continuous and strictly increasing, then $c(\mathbf{w}, y)$ is:*

1. zero when $y = 0$,
2. continuous on its domain,
3. for all $\mathbf{w} \gg 0$, strictly increasing and unbounded above in y ,
4. increasing in $\mathbf{w}$,
5. homogeneous of degree one in $\mathbf{w}$,
6. concave in $\mathbf{w}$.

These properties have straightforward economic interpretations.

1. The condition $c(\mathbf{w}, 0) = 0$ means that with no output target, there is no variable cost.
2. The fact that cost is increasing and unbounded in y means that producing more output requires more inputs, and those additional inputs cost more.
3. The fact that cost is increasing in $\mathbf{w}$ means that higher input prices raise production cost.
4. Homogeneity of degree one in $\mathbf{w}$ means that if all input prices double, minimum cost also doubles.
5. Finally, concavity in $\mathbf{w}$ reflects the firm's ability to reoptimize its input mix when relative prices change, shifting toward relatively cheap inputs whenever possible.

Lemma 6.2.8 (Shephard's Lemma). *If f is strictly quasiconcave, then*

$$\frac{\partial c(\mathbf{w}^0, y^0)}{\partial w_i} = x_i(\mathbf{w}^0, y^0), \quad i = 1, \dots, n.$$

This means that the derivative of the cost function with respect to the price of input i recovers the firm's conditional demand for that input. In economic terms, if the price of input i rises slightly, the resulting increase in minimum cost is exactly equal to the amount of input i the firm uses at the cost-minimizing bundle. This is the exact production-side analogue of Shephard's lemma from expenditure minimization in consumer theory.

6.3 Duality in Production

Cost minimization is the production-side analogue of expenditure minimization. For this reason, production and cost are dual objects, just as utility and expenditure are dual on the consumer side.

The key idea is to start with a production function, derive its cost function, and then recover production information back from the cost function. This means that technology and cost are two different representations of the same underlying economic structure.

If $c : \mathbb{R}_{++}^n \times \mathbb{R}_+ \rightarrow \mathbb{R}_+$ satisfies the cost-function properties, define

$$f(\mathbf{x}) = \max\{y \geq 0 : \mathbf{w} \cdot \mathbf{x} \geq c(\mathbf{w}, y) \text{ for all } \mathbf{w} \gg 0\}.$$

The recovered function f is increasing, unbounded above, and quasiconcave, and the cost function generated by this f is exactly the original function c . This shows that any function with the right cost-function properties can be interpreted as arising from some production technology.

- If the original production function was quasiconcave, then this recovery procedure returns the original technology.
- If the original production function was not quasiconcave, then the recovered object is instead a quasiconcave version of it.

Thus, the cost function preserves the part of technology that is consistent with well-behaved cost minimization.

Duality matters because it is sometimes easier to estimate cost than to measure production technology directly. Once a cost function has been estimated, one can recover production implications from it. This mirrors the logic on the consumer side, where value and expenditure functions can be used to recover information about preferences. More generally, any function with the correct cost-function properties corresponds to some production function, so the dual approach is both theoretically powerful and practically useful.

6.4 Short Run and Long Run

- In the short run, some inputs are fixed. When some inputs are fixed, the firm faces short-run restricted cost functions.
- In the long run, all inputs can vary, and entry and exit may also be possible. When no inputs are fixed, the firm faces long-run cost functions.

Definition 6.4.1. Starting from the total cost function $c(\mathbf{w}, y)$, define

$$AC(y) = \frac{c(\mathbf{w}, y)}{y}, \quad MC(y) = \frac{\partial c(\mathbf{w}, y)}{\partial y}.$$

Here, average cost is the cost per unit of output, while marginal cost is the cost of producing one more unit of output.

Example 6.4.2 (Cobb–Douglas Example). If

$$c(\mathbf{w}, y) = Ay^r,$$

then

$$AC(y) = Ay^{r-1}, \quad MC(y) = Ary^{r-1}.$$

This shows how the exponent r determines the shape of the cost curves. If $r = 1$, both average cost and marginal cost are constant. If $r > 1$, marginal cost is increasing. If $r < 1$, marginal cost is declining, which leads to the same competitive-profit problems discussed earlier. In particular, declining marginal cost gives firms an incentive to expand output too aggressively, and a finite interior profit maximum may fail to exist.

In the unrestricted long-run problem studied earlier, there are no fixed costs built into the model. Fixed costs emerge when some inputs are constrained or fixed in the short run. In that case, total cost splits into fixed and variable components. A fixed cost can also arise from something like a licence fee or an entry fee that must be paid before production begins. Thus, fixed cost need not come only from technological rigidity; it can also come from institutional or contractual requirements.

7 Profit Maximization

7.1 Competitive Firms

The output price is $p \geq 0$. The input price vector is $\mathbf{w} = (w_1, \dots, w_n) \geq \mathbf{0}$. The firm takes p and $\mathbf{w}$ as given. It chooses output and inputs to maximize profit. In the competitive benchmark, a competitive firm does not choose prices; it chooses quantities given prices.

Definition 7.1.1 (The Firm's Problem). For a single-output technology f , the firm solves

$$\max_{(\mathbf{x}, y) \geq \mathbf{0}} py - \mathbf{w} \cdot \mathbf{x} \quad \text{s.t.} \quad f(\mathbf{x}) \geq y. \quad (3.6)$$

Revenue is py . Cost is $\mathbf{w} \cdot \mathbf{x}$. Feasibility requires the chosen inputs to be able to produce the chosen output.

Because the production function is strictly increasing, the feasibility constraint binds at optimum. Thus the firm's problem can be written as

$$\max_{(\mathbf{x}, y) \geq \mathbf{0}} py - \mathbf{w} \cdot \mathbf{x} \quad \text{s.t.} \quad f(\mathbf{x}) = y.$$

If $f(\mathbf{x}) > y$, the firm is using enough inputs to produce more output than it actually sells. It can reduce some inputs, lower cost, and raise profit.

Substituting $y = f(\mathbf{x})$ gives

$$\max_{\mathbf{x} \in \mathbb{R}_+^n} pf(\mathbf{x}) - \mathbf{w} \cdot \mathbf{x}. \quad (3.7)$$

This is often the cleanest formulation when the production function is explicit. It also makes the first-order conditions transparent.

Assume an interior optimum $\mathbf{x}^* \gg \mathbf{0}$. Then the gradient of the maximand must vanish:

$$p \frac{\partial f(\mathbf{x}^*)}{\partial x_i} = w_i, \quad i = 1, \dots, n.$$

The left-hand side is the marginal revenue product of input i . The right-hand side is the unit cost of input i .

If all input prices are positive, the first-order conditions imply

$$\frac{\partial f(\mathbf{x}^*)/\partial x_i}{\partial f(\mathbf{x}^*)/\partial x_j} = \frac{w_i}{w_j} \quad \text{for all } i, j. \quad (\text{B4B Equalization})$$

This is the same technical tradeoff condition as in cost minimization. The ratio of marginal products must match the ratio of input prices. This is the same MRTS condition inherited from cost minimization.

7.2 The Profit Function

When the profit maximization problem has a unique solution,

- optimal output $y^* = y(p, \mathbf{w})$ is the firm's output supply function,
- and optimal inputs $\mathbf{x}^* = \mathbf{x}(p, \mathbf{w})$ are the firm's input demand functions.

An important distinction is that conditional input demands depend on $(\mathbf{w}, y)$, while profit-maximizing input demands depend only on prices $(p, \mathbf{w})$.

Definition 7.2.1 (The Profit Function). The profit function is the maximum-value function

$$\pi(p, \mathbf{w}) = \max_{(\mathbf{x}, y) \geq \mathbf{0}} py - \mathbf{w} \cdot \mathbf{x} \quad \text{s.t.} \quad f(\mathbf{x}) \geq y.$$

It depends only on prices. It summarizes the value of the firm's optimization problem.

There are two equivalent ways to solve profit maximization.

1. Solve directly using the production function:

$$\max_{\mathbf{x}} pf(\mathbf{x}) - \mathbf{w} \cdot \mathbf{x}.$$

2. Or solve in two steps:

$$\max_{y \geq 0} py - c(\mathbf{w}, y). \quad (3.8)$$

Interpretation: first minimize cost for each target output. Then choose the output level that maximizes revenue minus cost.

If $y^* > 0$ solves

$$\max_{y \geq 0} py - c(\mathbf{w}, y),$$

then the first-order condition is

$$p - \frac{dc(\mathbf{w}, y^*)}{dy} = 0.$$

So at an interior optimum,

$$p = MC(y^*).$$

For the interior solution above to be a maximum, we need

$$\frac{d^2 c(\mathbf{w}, y^*)}{dy^2} \geq 0.$$

Marginal cost should be non-decreasing at the optimum. This is why decreasing marginal cost creates trouble for competitive profit maximization.

With increasing returns, scaling up production can keep raising profit. With constant returns, zero profit may leave the scale of operation indeterminate. This is why decreasing or, with care, constant returns are central to the competitive benchmark.

Theorem 7.2.2 (Properties of the Profit Function). *If f satisfies Assumption 3.1, then where well-defined, $\pi(p, \mathbf{w})$ is continuous and:*

1. *increasing in p ,*
2. *decreasing in $\mathbf{w}$,*
3. *homogeneous of degree one in $(p, \mathbf{w})$,*
4. *convex in $(p, \mathbf{w})$.*

Theorem 7.2.3 (Supply and Input Demand). *Suppose f is strictly concave and π is twice continuously differentiable. Then:*

1. *Homogeneity of degree zero:*

$$y(tp, t\mathbf{w}) = y(p, \mathbf{w}), \quad x_i(tp, t\mathbf{w}) = x_i(p, \mathbf{w}).$$

2. *Own-price effects:*

$$\frac{\partial y}{\partial p} \geq 0, \quad \frac{\partial x_i}{\partial w_i} \leq 0.$$

Example 7.2.4 (Cobb–Douglas Example). Consider

$$y = f(\mathbf{x}) = x_1^\alpha x_2^\beta, \quad \alpha, \beta > 0, \quad \alpha + \beta < 1.$$

The condition $\alpha + \beta < 1$ gives strict concavity on $\mathbb{R}_{++}^2$. More generally, Cobb–Douglas is quasiconcave on $\mathbb{R}_+^2$ and strictly quasiconcave on $\mathbb{R}_{++}^2$.

Substitute the production function into profits:

$$\pi(\mathbf{x}) = px_1^\alpha x_2^\beta - (w_1 x_1 + w_2 x_2).$$

Because the objective is differentiable and strictly concave, we can use the first-order conditions to characterize the unique optimum.

The first-order conditions are

$$p\alpha x_1^{\alpha-1} x_2^\beta = w_1, \quad p\beta x_1^\alpha x_2^{\beta-1} = w_2.$$

Because $y = x_1^\alpha x_2^\beta$, we can rewrite the FOCs as

$$x_1 = \frac{\alpha p}{w_1} y, \quad x_2 = \frac{\beta p}{w_2} y. \tag{4}$$

Substitute (4) back into $y = x_1^\alpha x_2^\beta$ to obtain

$$y = \left(\frac{\alpha p}{w_1} y\right)^\alpha \left(\frac{\beta p}{w_2} y\right)^\beta.$$

Collecting terms,

$$y^{1-(\alpha+\beta)} = \left(\frac{\alpha p}{w_1}\right)^\alpha \left(\frac{\beta p}{w_2}\right)^\beta.$$

Therefore,

$$y(p, \mathbf{w}) = \left[\left(\frac{\alpha}{w_1}\right)^\alpha \left(\frac{\beta}{w_2}\right)^\beta p^{\alpha+\beta} \right]^{\frac{1}{1-(\alpha+\beta)}}. \quad (5)$$

Output rises with the output price. Output falls when either input price rises.

Use (4) together with (5):

$$x_1(p, \mathbf{w}) = \left[\left(\frac{\alpha}{w_1}\right)^{1-\beta} \left(\frac{\beta}{w_2}\right)^\beta p \right]^{\frac{1}{1-(\alpha+\beta)}}, \quad x_2(p, \mathbf{w}) = \left[\left(\frac{\alpha}{w_1}\right)^\alpha \left(\frac{\beta}{w_2}\right)^{1-\alpha} p \right]^{\frac{1}{1-(\alpha+\beta)}}.$$

Using

$$x_1 = \frac{\alpha p}{w_1} y, \quad x_2 = \frac{\beta p}{w_2} y,$$

we get

$$\pi(p, \mathbf{w}) = py - (w_1 x_1 + w_2 x_2) = py - (\alpha py + \beta py).$$

So

$$\pi(p, \mathbf{w}) = (1 - (\alpha + \beta))py.$$

Substituting $y(p, \mathbf{w})$ into the previous expression yields

$$\pi(p, \mathbf{w}) = (1 - (\alpha + \beta))p \left[\left(\frac{\alpha}{w_1}\right)^\alpha \left(\frac{\beta}{w_2}\right)^\beta p^{\alpha+\beta} \right]^{\frac{1}{1-(\alpha+\beta)}}.$$

Profit is increasing in p but decreasing in $\mathbf{w}$.

Suppose

$$w_1 = 1, \quad w_2 = 2, \quad p = 10, \quad \alpha = \frac{1}{2}, \quad \beta = \frac{1}{4}.$$

Then

$$\pi(10, \mathbf{w}) = 78.125.$$

If we double all prices,

$$\pi(20, 2\mathbf{w}) = 2\pi(10, \mathbf{w}) = 156.25.$$

This illustrates homogeneity of degree one of the profit function in $(p, \mathbf{w})$.

If $\alpha + \beta = \frac{1}{2}$, then

$$y(p, \mathbf{w}) = kp$$

for some constant $k > 0$ depending on $\alpha, \beta, \mathbf{w}$.

Output supply is linear through the origin. The supply curve is the marginal cost curve ($p = MC(y)$). So this corresponds to linear marginal cost.

When $\alpha + \beta = \frac{1}{2}$, the previous lecture showed that the cost function has the form

$$c(\mathbf{w}, y) = Ay^2$$

for some constant $A > 0$.

Quadratic total cost implies linear marginal cost. Linear marginal cost implies linear supply under competition. The production, cost, and supply representations are all connected.

What if $\alpha + \beta \geq 1$? The direct profit-maximization formulas above relied on $\alpha + \beta < 1$, so they no longer apply.

But the cost function is still available because its derivation only used strict quasiconcavity. That lets us study profit maximization via

$$\max_{y \geq 0} py - c(\mathbf{w}, y).$$

- Case 1: Constant Returns $\alpha + \beta = 1$. Then the cost function is linear:

$$c(\mathbf{w}, y) = dy$$

for some constant $d > 0$. Hence

$$\pi(y) = py - dy = (p - d)y.$$

1. If $p > d$, profit rises with y : no finite maximizer.
 2. If $p < d$, every positive output makes losses: the unique solution is $y = 0$ and $\mathbf{x} = \mathbf{0}$.
 3. If $p = d$, profit is zero for every scale: the scale of the firm is indeterminate.
- Case 2: Increasing Returns $\alpha + \beta > 1$. Now the cost function takes the form

$$c(\mathbf{w}, y) = dy^\delta, \quad \delta = \frac{1}{\alpha + \beta} < 1.$$

Total cost is concave in output. Therefore marginal cost is declining.

Why increasing returns break the competitive solution? If

$$c(\mathbf{w}, y) = dy^\delta, \quad 0 < \delta < 1,$$

then $MC(y) = d\delta y^{\delta-1}$ falls with y .

- Profit initially may fall, but eventually rises without bound.
- So there is no profit-maximizing finite output.
- A competitive firm facing decreasing marginal cost wants to keep expanding.

Example 7.2.5 (Concrete Counterexample). Suppose:

$$p = \alpha = \beta = w_1 = w_2 = 1.$$

Then $\alpha + \beta = 2 > 1$, so $\delta = \frac{1}{2}$ and

$$c(\mathbf{w}, y) = 2y^{1/2}.$$

Profit is

$$\pi(y) = 2y - 2y^{1/2}.$$

- It starts at zero.
- It falls below zero for small positive y .
- After $y = \frac{1}{4}$, it rises without bound.

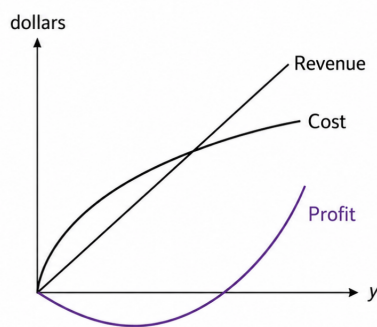


Figure 4: Nonexistence Under Increasing Returns: revenue is linear, cost is concave, and profit ultimately becomes unbounded above.

7.3 Short Run and Welfare

In the short run, some inputs are fixed and cannot be adjusted. This creates a restricted profit function. Short-run profit is typically lower than long-run profit because the firm has fewer degrees of freedom. Fixed or startup costs become visible in this formulation. Fixing an input acts like introducing a fixed-cost component.

Example 7.3.1.

$$f(\mathbf{x}) = x_1^{1/2} x_2^{1/4}, \quad x_2 = 16, \quad w_2 = 2.$$

The short-run production function becomes

$$y = 2x_1^{1/2}.$$

Profit is

$$\pi(x_1) = 2px_1^{1/2} - w_1x_1 - 32.$$

The first-order condition gives

$$x_1(p, \mathbf{w}) = \frac{p^2}{w_1^2}, \quad y(p, \mathbf{w}) = \frac{2p}{w_1}.$$

So short-run profit is

$$\pi(p, \mathbf{w}) = \frac{p^2}{w_1} - 32.$$

With $w_1 = 1$ and $p = 10$, short-run profit is $100 - 32 = 68$, below the long-run value 78.125 calculated above.

For producers, welfare is simply profit. We do not need a separate willingness-to-pay construction. Everything is already measured in dollars.

For consumers we had to recover welfare indirectly from preferences, utility, expenditure, and demand. For firms under profit maximization, welfare is directly observed as profit.

Part II

Markets and Welfare

8 Perfect Competition

8.1 Competitive Market Setup

Definition 8.1.1 (Perfectly Competitive Markets). Buyers and sellers take prices as given. Market demand is the sum of individual demands. Market supply is the sum of individual supplies.

Behind this sit utility maximization by buyers and profit maximization by sellers.

In the short run, some inputs are fixed and entry/exit is not allowed. In the long run, all inputs are variable and entry/exit is allowed.

Example 8.1.2 (Market Environment). Demand side: There are 64 consumers, each with utility

$$u(y, z) = \ln y + z,$$

with the price of z fixed at $p_z = 1$.

Supply side: There are 4 sellers, each with production function

$$y = f(x_1, x_2) = x_1^{1/4} x_2^{1/4},$$

and input prices fixed at $w_1 = 4$ and $w_2 = 1$.

In the short run, we fix $x_2 = 16$. In the long run, all inputs may vary and the number of sellers may also vary.

We denote the price of the market good y by p .

Definition 8.1.3 (Short-Run Competitive Equilibrium). A perfectly competitive equilibrium is a price p and quantity Y such that market demand equals market supply.

Definition 8.1.4 (Long-Run Competitive Equilibrium). A long-run competitive equilibrium is a price p , quantity Y , and number of firms n such that:

1. Market demand equals market supply.
2. Profit of each seller is zero.

For each consumer, the quasi-linear utility

$$u(y, z) = \ln y + z, \quad \text{with } p_z = 1.$$

Utility maximization implies the interior demand for y

$$y^d(p) = \frac{1}{p}.$$

This is a quasi-linear demand. Therefore, there are no income effects for good y , provided income is high enough.

The expenditure on that demand is

$$py^d(p) = p \cdot \frac{1}{p} = 1.$$

So consumers need income at least 1 to afford the interior choice. We assume this is the case going forward.

There are 64 identical consumers, so market demand is simply

$$Y^D(p) = 64y^d(p) = \frac{64}{p}.$$

This is the market demand curve for the example economy. Identical consumers make aggregation easy here.

In general, consumers need not all be identical.

Example 8.1.5. If extra consumers had Cobb–Douglas preferences, each would contribute a demand of the form

$$y = \frac{A}{p}$$

for some constant $A > 0$.

Market demand is just the horizontal sum of individual demands, heterogeneous or homogeneous.

8.2 Short Run Competition

With

$$y = x_1^{1/4} x_2^{1/4} \quad \text{and} \quad x_2 = 16,$$

we obtain the short-run production function

$$y = 2x_1^{1/4}.$$

Output depends only on the variable input x_1 . The input x_2 generates a fixed cost in the short run. There are two equivalent methods.

1. Two-step method:

- derive the cost function $C(y)$,
- then maximize

$$\pi(y) = py - C(y).$$

2. One-step method: maximize profit directly in inputs and output.

We follow the two-step method here.

Because input markets are competitive, w_1 and w_2 are constant:

$$C(y) = w_1x_1 + w_2x_2.$$

With $x_2 = 16$, the fixed-cost component is

$$w_2x_2 = 1 \cdot 16 = 16.$$

So

$$C(y) = 4x_1(y) + 16.$$

We now need x_1 as a function of output y .

Since

$$y = 2x_1^{1/4}$$

is strictly increasing in x_1 , we can invert it:

$$x_1(y) = \left(\frac{y}{2}\right)^4 = \frac{y^4}{16}.$$

This is the short-run input demand for x_1 needed to produce output y .

Substitute $x_1(y) = \frac{y^4}{16}$ into the cost expression:

$$C(y) = 4\left(\frac{y^4}{16}\right) + 16 = \frac{y^4}{4} + 16.$$

Variable cost: $\frac{y^4}{4}$. Fixed cost: 16.

Profit is revenue minus cost:

$$\pi(y) = py - C(y) = py - \frac{y^4}{4} - 16.$$

Revenue is linear in output.

Cost is strictly convex in output, so $-C(y)$ is strictly concave. Therefore profit is strictly concave.

We can check that

$$\pi''(y) = -3y^2 < 0$$

when $y > 0$. So π is strictly concave in this region.

Therefore, any point where

$$\frac{d\pi}{dy} = 0$$

is the unique global maximizer. This is the standard $MR = MC$ logic.

Differentiate the short-run profit function:

$$\frac{d\pi}{dy} = p - y^3.$$

The first-order condition gives

$$y^s(p) = p^{1/3}.$$

This is one seller's short-run supply function.

If we solve directly in the variable input,

$$\pi(x_1) = p2x_1^{1/4} - 4x_1 - 16,$$

we will get the same answer.

To check this, differentiate with respect to x_1 :

$$\frac{d\pi}{dx_1} = \frac{p}{2}x_1^{-3/4} - 4.$$

Set the first-order condition equal to zero:

$$\frac{p}{2}x_1^{-3/4} - 4 = 0.$$

Then

$$x_1(p) = \left(\frac{p}{8}\right)^{4/3} = \frac{p^{4/3}}{16}.$$

Using the short-run production function,

$$y = 2x_1^{1/4},$$

we get

$$y^s(p) = p^{1/3}.$$

Thus, the direct method gives the same short-run supply function.

Plug the profit-maximizing output into $x_1(y) = \frac{y^4}{16}$. Then

$$x_1(p) = \frac{(p^{1/3})^4}{16} = \frac{p^{4/3}}{16}.$$

So the firm's variable-input demand rises with the output price.

There are 4 identical sellers, each supplying $p^{1/3}$. Therefore, market supply is

$$Y^S(p) = 4p^{1/3}.$$

This is the horizontal sum of the firms' supply curves.

Demand is

$$Y^D = \frac{64}{p}.$$

Supply is

$$Y^S = 4p^{1/3}.$$

Their intersection gives the short-run competitive equilibrium.

In the diagram, the vertical axis is p , and the horizontal axis is Y . The downward-sloping curve is Y^D , the upward-sloping curve is Y^S , and their intersection is the short-run equilibrium.

Solve

$$Y^D(p) = Y^S(p)$$

which means

$$\frac{64}{p} = 4p^{1/3} \Rightarrow p = 8.$$

Plugging back in,

$$Y = 8.$$

At the short-run equilibrium price $p = 8$, each seller supplies

$$y^s = 8^{1/3} = 2.$$

Each consumer buys

$$y^d = \frac{1}{8}.$$

The 64 consumers together buy

$$64 \cdot \frac{1}{8} = 8,$$

matching total supply.

For each seller,

$$\pi(y) = py - \frac{y^4}{4} - 16.$$

At $p = 8$ and $y = 2$, we have

$$\pi = 8 \cdot 2 - \frac{2^4}{4} - 16 = -4.$$

This is still better than shutting down if the fixed cost must be paid anyway, because shutdown would leave profit of -16 .

8.3 Long Run Competition

In the long run:

- both inputs x_1 and x_2 are variable,
- the number of sellers is variable,
- we assume a fixed cost of entry $F = 4$.

Recall. The variable cost for this Cobb–Douglas technology with $\alpha = \beta = \frac{1}{4}$ and input prices $w_1 = 4$, $w_2 = 1$ simplifies to

$$c(y) = 4y^2.$$

Adding the fixed cost of entry gives the long-run total cost of each seller:

$$C(y) = 4 + 4y^2.$$

Profit is now

$$\pi(y) = py - C(y) = py - (4 + 4y^2).$$

This is concave in y . So we can again use the first-order approach.

For a positive-output competitive equilibrium, we need:

1. Profit maximization:

$$p = MC.$$

2. Zero profit:

$$TR = TC \Rightarrow p = AC.$$

Hence, we can search for output where

$$MC = AC.$$

For $C(y) = 4 + 4y^2$, we have

$$MC(y) = C'(y) = 8y,$$

and

$$AC(y) = \frac{C(y)}{y} = \frac{4}{y} + 4y.$$

Set

$$8y = \frac{4}{y} + 4y.$$

This implies

$$4y = \frac{4}{y} \Rightarrow y = 1.$$

So the long-run equilibrium output of each seller is

$$y = 1.$$

At the profit-maximizing output $y = 1$, we have

$$p = MC(1) = 8.$$

You can also check

$$AC(1) = \frac{4}{1} + 4(1) = 8.$$

So zero profit and profit maximization are both satisfied at

$$p = 8, \quad y = 1.$$

At price $p = 8$, market demand is

$$Y = \frac{64}{8} = 8.$$

Since each seller supplies one unit in the long run, we need $n = 8$ sellers.

8.4 Edge Cases

Question 1: What If There Were No Entry Cost?

Without the entry cost, the cost function would be

$$C(y) = 4y^2.$$

Then

$$AC(y) = 4y, \quad MC(y) = 8y.$$

The condition $AC = MC$ gives

$$4y = 8y \Rightarrow y = 0.$$

If $y = 0$, then competitive pricing would require

$$p = 0.$$

But market demand

$$Y^D = \frac{64}{p}$$

is not well-defined at $p = 0$. The limit pushes toward infinitely many firms, each producing an infinitesimally small amount.

Question 2: What If the Number of Sellers Is Not an Integer?

If the number of sellers is not an integer, choose the largest integer n that still yields positive profit, then let price adjust to clear the market given that number of firms.

Idea: Each seller may earn a small positive profit, but entry of one more firm would drive profit below zero.

Question 3: What If Sellers Differ?

If there are different seller types:

- firms with inferior technologies or higher average costs tend to be driven out,
- the market may end up with a mix of seller types,
- long-run selection works through entry and exit.

Question 4: Can There Be Multiple Long-Run Equilibria?

Yes. The textbook provides an example with constant returns to scale. Under constant returns, zero-profit conditions can leave firm scale or market structure weakly pinned down.

Remark. Competitive equilibrium can be unique in some examples and non-unique in others.

9 Imperfect Competition

9.1 Monopoly

Main Difference: Under market power, the seller does not take price as given. So we no longer look for a price where supply equals demand.

A monopoly typically involves barriers to entry such as:

- large startup costs,
- government restrictions,
- patents or exclusive control of a technology.

The seller now chooses price as well as output and inputs. Consumers still take the chosen price as given and move along their demand curve. More generally, a seller with market power may also have the power to use more complicated pricing rules such as price discrimination.

Example 9.1.1. Take

$$C(y) = 4 + 4y^2$$

and market demand

$$Y^D = \frac{64}{p}.$$

The monopolist recognizes that it can choose output y and then choose the price that places consumers on the demand curve, which is the highest price consumers will pay for that level of output.

From

$$y = \frac{64}{p},$$

we obtain the inverse demand curve

$$p(y) = \frac{64}{y}.$$

So profit can be written purely as a function of output:

$$\pi(y) = p(y)y - C(y).$$

Substitute inverse demand into profit:

$$\pi(y) = \frac{64}{y} \cdot y - (4 + 4y^2) = 64 - 4 - 4y^2.$$

So

$$\pi(y) = 60 - 4y^2.$$

Because revenue is constant,

$$MR = 0.$$

There is no economically meaningful interior solution. Also, $y = 0$ is not admissible because it would imply an undefined or infinite price.

What the monopolist would like to do is:

- set a very high price,
- restrict output drastically,
- approach the highest achievable profit as a limit.

In reality, demand often has a finite choke price. Then the monopolist chooses that maximum feasible price and the corresponding quantity.

Example 9.1.2. Now instead use linear demand:

$$Y^D = 40 - \frac{p}{2}.$$

Equivalently, inverse demand is

$$p(y) = 80 - 2y.$$

Keep the same cost function:

$$C(y) = 4 + 4y^2.$$

Profit is

$$\pi(y) = p(y)y - C(y) = (80 - 2y)y - (4 + 4y^2).$$

Simplifying,

$$\pi(y) = 80y - 6y^2 - 4.$$

Differentiate:

$$\pi'(y) = 80 - 12y.$$

The first-order condition gives

$$y^* = \frac{80}{12} = \frac{20}{3} \approx 6.67.$$

The second derivative is

$$\pi''(y) = -12 < 0,$$

so this is the unique profit-maximizing output.

Plug the optimal output into inverse demand:

$$p^* = 80 - 2\left(\frac{20}{3}\right) = \frac{200}{3} \approx 66.67.$$

The monopolist chooses both price and quantity jointly. This is fundamentally different from perfect competition.

9.2 Oligopoly

Definition 9.2.1. An oligopoly is a market structure with more than one seller, but not so many sellers that perfect competition is a good approximation.

Like a monopoly, each seller has market power. There are many models of oligopoly.

In an oligopoly, one seller's choice affects the profitability of the others. If firm 1 raises output, the market price changes. That price change also changes firm 2's profit. Therefore, each firm's optimization problem depends on what its rivals do. This is why oligopoly is naturally analyzed using game theory.

9.2.1 Cournot

Two Sellers Model: Duopoly

Example 9.2.2 (Duopoly Environment). Consider a market with two sellers.

Demand:

$$D(p) = 100 - 2p$$

Costs: for seller $i = 1, 2$,

$$C_i(q_i) = q_i^2.$$

The strategic variables are quantities. The question is: what happens in equilibrium?

Because firms choose quantities and demand is monotone, it is convenient to invert the demand curve.

$$q = 100 - 2p \quad \Rightarrow \quad p(q) = 50 - \frac{q}{2}.$$

Total market quantity is

$$q = q_1 + q_2$$

So price depends on the sum of both firms' outputs.

Each firm's profit is revenue minus cost:

$$\pi_1(q_1, q_2) = p(q_1 + q_2)q_1 - C_1(q_1)$$

$$\pi_2(q_1, q_2) = p(q_1 + q_2)q_2 - C_2(q_2)$$

Substituting inverse demand and costs gives

$$\pi_1(q_1, q_2) = \left(50 - \frac{q_1 + q_2}{2}\right) q_1 - q_1^2$$

$$\pi_2(q_1, q_2) = \left(50 - \frac{q_1 + q_2}{2}\right) q_2 - q_2^2$$

Thus, each firm's profit depends on its own quantity and the other firm's quantity through the market price.

The standard equilibrium concept in games is a Nash equilibrium.

Definition 9.2.3. In an n -player game, a Nash equilibrium is an n -tuple of strategies

$$\mathbf{s} = (s_1, \dots, s_n)$$

such that each player's strategy maximizes that player's utility given the strategies of the others.

In quantity-setting oligopoly, the strategies are quantities. In richer models, strategies might include prices, advertising, technology, lobbying, or information acquisition.

Definition 9.2.4. In a duopoly, a Nash equilibrium is a pair (q_1, q_2) such that each firm's quantity maximizes its own profit given the quantity chosen by the other firm. A pair (q_1, q_2) is a Nash equilibrium if

$$\pi_1(q_1, q_2) \geq \pi_1(q'_1, q_2) \quad \text{for all } q'_1 \geq 0$$

and

$$\pi_2(q_1, q_2) \geq \pi_2(q_1, q'_2) \quad \text{for all } q'_2 \geq 0.$$

With more sellers, we simply get one such profit-maximization condition for each seller.

Definition 9.2.5 (Best Responses). A firm's best response is the profit-maximizing choice given rivals' actions.

Fix q_2 . Firm 1 solves a one-variable profit maximization problem in q_1 .

Fix q_1 . Firm 2 solves a one-variable profit maximization problem in q_2 .

Then a Nash equilibrium is a point where both firms are simultaneously giving best responses.

A useful mathematical perspective: Collect all firms' best responses into a best-response correspondence. A Nash equilibrium is a fixed point of that correspondence. Existence proofs often rely on fixed-point theorems.

This mirrors the way competitive equilibrium existence is also often established using fixed-point arguments.

For firm 1,

$$\pi_1(q_1, q_2) = \left(50 - \frac{q_1 + q_2}{2}\right) q_1 - q_1^2$$

Differentiate with respect to q_1 :

$$\frac{\partial \pi_1}{\partial q_1} = 50 - \frac{q_1 + q_2}{2} - \frac{q_1}{2} - 2q_1$$

Setting this equal to zero gives firm 1's FOC:

$$50 - \frac{q_1 + q_2}{2} - \frac{q_1}{2} - 2q_1 = 0$$

The second derivative is

$$\frac{\partial^2 \pi_1}{\partial q_1^2} = -\frac{1}{2} - \frac{1}{2} - 2 < 0$$

So firm 1's profit is strictly concave in its own strategy q_1 .

Therefore, any point satisfying the FOC is the unique best response to a given q_2 .

Solving firm 1's FOC for q_1 yields

$$q_1(q_2) = \frac{100 - q_2}{6}$$

This is firm 1's best response function.

By symmetry,

$$q_2(q_1) = \frac{100 - q_1}{6}$$

So both firms have downward-sloping best responses.

If the rival produces more, your best response is to produce less.

The equilibrium is at the intersection of the two best-response lines. Each line tells one firm's optimal quantity for each quantity chosen by the other.

The equilibrium must solve

$$q_1 = \frac{100 - q_2}{6}, \quad q_2 = \frac{100 - q_1}{6}.$$

This system has the unique solution

$$q_1 = q_2 = \frac{100}{7}.$$

So the duopoly has a unique interior Nash equilibrium:

$$\left(\frac{100}{7}, \frac{100}{7}\right).$$

How do we rule out boundary equilibria? Suppose

$$q_1 = 0.$$

Then firm 2's best response is

$$q_2 = \frac{100}{6}.$$

But then firm 1's best response to that is

$$q_1 = \frac{100 - 100/6}{6} > 0.$$

So a boundary point like $q_1 = 0$ cannot be a Nash equilibrium.

Total output at the Nash equilibrium is

$$q = q_1 + q_2 = \frac{200}{7}.$$

So market price is

$$p = 50 - \frac{q}{2} = 50 - \frac{100}{7} = \frac{250}{7}.$$

Oligopoly output is below the competitive benchmark. Price is above marginal-cost pricing.

When the model is smooth, the procedure can be summarized as follows:

1. Write down each seller's profit function.
2. Check concavity in that seller's own strategy.
3. Use first-order conditions.
4. Solve the FOCs simultaneously.
5. Check for boundary solutions that are not captured by the FOCs.

In simple examples, existence holds. Concave demand and convex costs tend to make profit concave, which usually leads to equilibrium existence.

In more general games, equilibrium may fail to exist.

Example 9.2.6. For example:

$$MC = 0, \quad P = \frac{1}{Q}.$$

Existence theorems typically rely on fixed-point methods.

General N-Seller Cournot

Now allow $n \geq 1$ sellers. For seller i , profit is

$$\pi_i(\mathbf{q}) = (a - bq) q_i - \frac{c}{2} q_i^2, \quad a, b, c > 0$$

where total output is

$$q = \sum_{j=1}^n q_j.$$

This nests the duopoly example in a more general linear-demand quadratic-cost environment.

Differentiate with respect to q_i :

$$a - bq - bq_i - cq_i = 0$$

The second derivative is

$$\frac{\partial^2 \pi_i}{\partial q_i^2} = -b - b - c < 0$$

So each seller's profit is strictly concave in its own quantity.

Rearranging the first-order condition gives

$$q_i(q) = \frac{a - bq}{b + c}$$

When expressed in terms of total output q , all firms have the same best-response formula in the symmetric case.

Sum the best-response equations over all i :

$$q = \frac{n(a - bq)}{b + c}$$

Solving for q gives the unique interior Nash-equilibrium total output:

$$q = \frac{na}{(n + 1)b + c}$$

Substituting back in, each seller produces

$$q_i = \frac{a}{(n + 1)b + c}$$

The equilibrium price is

$$p = \frac{a(b + c)}{(n + 1)b + c}$$

Each seller's profit is

$$\pi_i = \frac{a^2(2b + c)}{2[(n + 1)b + c]^2}$$

As the number of sellers grows,

$$q_i \rightarrow 0$$

Each individual firm's quantity shrinks toward zero. The market begins to resemble perfect competition.

Long Run with Entry

As with perfect competition, we assume that in the long run:

- Entry occurs while profits are positive.
- Exit occurs if profits are negative.
- In practice, choose the largest integer n such that each active firm earns non-negative (or positive) profit; entry of one more firm would drive profit below zero.

Suppose each seller also pays a fixed entry cost $F > 0$. Then long-run profit per seller becomes

$$\pi_i = \frac{a^2(2b + c)}{2[(n + 1)b + c]^2} - F$$

The long-run condition is to find where profit is driven to zero.

Set

$$\frac{a^2(2b + c)}{2[(n + 1)b + c]^2} - F = 0$$

Solving for n gives

$$n = \frac{1}{b} \left(a \sqrt{\frac{2b+c}{2F}} - c \right) - 1$$

This treats sellers as a continuous quantity.

Using

$$a = 100, \quad b = 1, \quad c = 2, \quad F = 9$$

we obtain

$$n \approx 46.14$$

So the practical long-run solution is

$$n = 46$$

Under oligopoly, each entrant imposes a strategic externality on incumbent firms.

Entry decisions therefore interact with best responses and profits in a way they do not under simple price taking.

This typically produces less entry than under a perfectly competitive benchmark with otherwise similar primitives.

But it could also lead to “too much” entry relative to the social optimum if entry does not increase total surplus but is still profitable through stealing business from other firms.

9.2.2 Bertrand Oligopoly

Cournot competition treats quantities as strategies. Bertrand competition treats prices as strategies.

Firms choose prices simultaneously. Consumers buy from the cheapest firm. A small change in the strategic variable produces a very different equilibrium.

Use the textbook’s simple homogeneous-good model.

Example 9.2.7. Two firms produce the same product. Each has constant marginal cost $c > 0$ and no fixed cost.

Market demand is

$$Q = \alpha - \beta p, \quad \alpha, \beta > 0$$

Each firm chooses a price $p_i \geq c$.

Consumers buy from the lowest-price firm.

$$q_1(p_1, p_2) = \begin{cases} \alpha - \beta p_1 & \text{if } p_1 < p_2, \\ \frac{1}{2}(\alpha - \beta p_1) & \text{if } p_1 = p_2, \\ 0 & \text{if } p_1 > p_2. \end{cases}$$

If one firm undercuts, it serves the whole market. If prices tie, firms split demand.

For prices below the choke price α/β , firm 1’s profit is

$$\pi_1(p_1, p_2) = \begin{cases} (p_1 - c)(\alpha - \beta p_1) & \text{if } c < p_1 < p_2, \\ \frac{1}{2}(p_1 - c)(\alpha - \beta p_1) & \text{if } c < p_1 = p_2, \\ 0 & \text{otherwise.} \end{cases}$$

The discontinuity is the reason first-order conditions are not the right tool here.

Firm 2's payoff is symmetric.

Suppose both firms set price equal to marginal cost:

$$p_1 = p_2 = c$$

Each firm serves half the market. Each unit is sold at cost, so each firm earns zero profit. Raising price gives the rival the whole market. Lowering price below c loses money on every unit. So neither firm can profitably deviate.

Suppose firm 1 sets $p_1 > c$. Firm 2 can earn positive profit by choosing a price just below p_1 . If firm 2 instead matched p_1 , it would split the market. By undercutting slightly, it gets almost the same margin and the whole market.

Thus firm 2's best response must satisfy

$$c < p_2 < p_1$$

By the same argument, if $p_2 > c$, firm 1's best response must satisfy

$$c < p_1 < p_2$$

Combining the two implications would require

$$p_2 < p_1 \quad \text{and} \quad p_1 < p_2$$

which is impossible.

Conclusion: The unique Bertrand equilibrium is

$$p_1 = p_2 = c$$

with zero profit for both firms.

The result is striking. There are only two firms. Each firm has market power in the sense that it is not a price taker. Yet equilibrium price is marginal cost, just like perfect competition.

Why this can fail: The result relies on homogeneous goods, identical costs, no capacity constraints, and pure price competition.

9.3 Monopolistic Competition

Cournot and Bertrand oligopoly in the textbook begin with homogeneous goods.

In monopolistic competition, many firms sell differentiated products. Products are close substitutes, but not perfect substitutes. Each firm faces a downward-sloping demand curve for its own variety. Entry from competitors reduces demand for your own variety.

Let $\mathbf{p} = (p_1, p_2, \dots)$ collect all variant prices. Demand for firm j 's product is

$$q_j = q_j(\mathbf{p})$$

with

$$\frac{\partial q_j}{\partial p_j} < 0, \quad \frac{\partial q_j}{\partial p_k} > 0 \quad \text{for } k \neq j.$$

Raising your own price lowers your demand. Raising a rival's price raises your demand.

Firm j 's profit is

$$\pi_j(\mathbf{p}) = p_j q_j(\mathbf{p}) - c_j(q_j(\mathbf{p}))$$

Profit depends on all prices through demand. In the short run, the active set of firms is fixed. In the long run, entry and exit are allowed.

In the short run, active firms choose prices to maximize profit given rivals' prices.

This is still a Nash equilibrium problem. If firm j chooses an interior price, the FOC can be arranged as

$$mr_j(q_j(\mathbf{p})) - mc_j(q_j(\mathbf{p})) = 0$$

Thus a monopolistic competitor chooses the point where marginal revenue equals marginal cost.

Short-run profit is not pinned down by the FOC alone. If demand is high relative to average cost, the firm earns positive profit. If demand is tangent to average cost, profit is zero. If demand is low relative to average cost, the firm can earn negative profit in the short run.

In the long run: Firms exit if profits are negative. New variants enter if existing firms can earn positive profit. Entry of close substitutes shifts each active firm's demand inward.

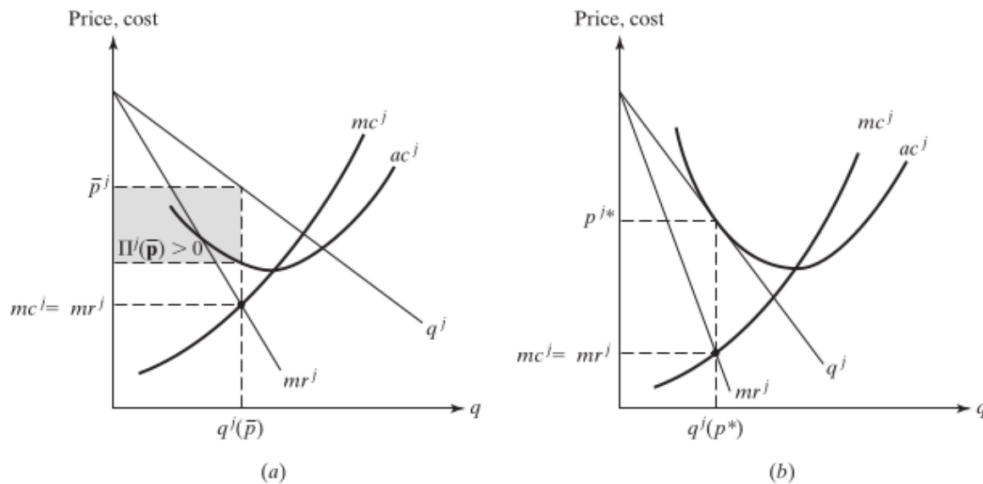


Figure 5: (a) Short-run and (b) long-run equilibrium in monopolistic competition.

The difference is mainly behavioral and institutional. In oligopoly, firms explicitly account for rivals' strategic responses. In monopolistic competition, each firm treats its residual demand curve as given. Product differentiation gives market power, but free entry removes long-run profit.

9.4 Extensions

The basic framework can be extended in many directions.

- Asymmetric costs.
- Asymmetric but interrelated demands.
- Goods that are not perfect substitutes.

- Asymmetric information.
- Nonlinear tariffs and other pricing rules.

Note: Nash equilibrium need not be socially efficient. Firms maximize private profit, not total welfare. Strategic interaction can generate outcomes that are poor relative to competitive benchmarks.

10 General Equilibrium Theory

10.1 Exchange Economy

Definition 10.1.1 (Exchange Economy). An exchange economy is written as

$$E = (u_i, \mathbf{e}^i)_{i=1}^I.$$

It consists of:

- consumers indexed by $i = 1, \dots, I$,
- n goods indexed by $k = 1, \dots, n$,
- for each consumer i , an endowment

$$\mathbf{e}^i = (e_1^i, \dots, e_n^i) \in \mathbb{R}_+^n,$$

- for each consumer i , a utility function u_i over bundles in $\mathbb{R}_+^n$.

Example 10.1.2. Consider Sarah with utility

$$u_1(x_1, x_2) = x_1^2 x_2$$

and endowment

$$\mathbf{e}^1 = (e_1^1, e_2^1) = (2, 2).$$

She can consume her endowment directly.

Alternatively, she can trade in competitive markets at prices

$$\mathbf{p} = (p_1, p_2).$$

In a pure exchange economy, Sarah's income is the value of her endowment:

$$m^1 = \mathbf{p} \cdot \mathbf{e}^1 = p_1 e_1^1 + p_2 e_2^1 = 2p_1 + 2p_2.$$

Her wealth depends on market prices. If we had production, she could also receive income from ownership shares in firms.

Given utility

$$u_1(x_1, x_2) = x_1^2 x_2,$$

consumer theory implies

$$x_1^1(\mathbf{p}, m^1) = \frac{2m^1}{3p_1}, \quad x_2^1(\mathbf{p}, m^1) = \frac{m^1}{3p_2}.$$

Substituting Sarah's endowment income $m^1 = \mathbf{p} \cdot \mathbf{e}^1 = 2p_1 + 2p_2$ gives

$$x_1^1(\mathbf{p}) = \frac{2(2p_1 + 2p_2)}{3p_1}, \quad x_2^1(\mathbf{p}) = \frac{2p_1 + 2p_2}{3p_2}.$$

When both prices scale up proportionally, income also scales up. **So only relative prices matter.**

Note: now subscripts are goods, as usual, and superscripts are consumer indices.

At prices $\mathbf{p} = (1, 2)$, Sarah's demands are

$$x_1 = \frac{2(2 + 4)}{3} = 4, \quad x_2 = \frac{2 + 4}{6} = 1.$$

Since her endowment is

$$\mathbf{e}^1 = (2, 2),$$

she buys 2 more units of good 1 and sells 1 unit of good 2.

Definition 10.1.3. For any good k , excess demand is defined as

$$z_k^i(\mathbf{p}) = x_k^i(\mathbf{p}, \mathbf{p} \cdot \mathbf{e}^i) - e_k^i.$$

Positive excess demand means the consumer buys from the market. Negative excess demand means the consumer supplies to the market.

At prices $\mathbf{p} = (1, 2)$,

$$z_1^1(\mathbf{p}) = 4 - 2 = 2,$$

so Sarah demands 2 extra units of good 1 from the market. Also,

$$z_2^1(\mathbf{p}) = 1 - 2 = -1,$$

so she supplies 1 unit of good 2 to the market.

If Sarah is the only consumer, then prices $\mathbf{p} = (1, 2)$ are not an equilibrium. She wants to buy more of good 1 than exists beyond her own endowment. She wants to sell some good 2.

Question: Are there prices at which her demand exactly matches her endowment, so that excess demand is zero in every market?

Market 1 clears if $x_1^1(\mathbf{p}) = 2$. That is,

$$\frac{2(2p_1 + 2p_2)}{3p_1} = 2.$$

Because only relative prices matter, normalize $p_1 = 1$. Then we solve for p_2 . With $p_1 = 1$, market 1 requires

$$\frac{2(2 + 2p_2)}{3} = 2.$$

This is solved by

$$p_2 = \frac{1}{2}.$$

At these prices, Sarah's demand for good 1 is exactly her endowment of good 1.

Now evaluate demand for good 2 at

$$\mathbf{p} = \left(1, \frac{1}{2}\right).$$

We get

$$x_2^1(\mathbf{p}) = \frac{2+1}{3(1/2)} = 2.$$

So Sarah's demand for good 2 also matches her endowment.

Conclusion: These prices form a general equilibrium in the one-consumer exchange economy.

Definition 10.1.4 (Allocation). An allocation is a vector $\mathbf{x} = (\mathbf{x}^1, \dots, \mathbf{x}^I)$, where each $\mathbf{x}^i = (x_1^i, \dots, x_n^i) \in \mathbb{R}_+^n$ is a bundle assigned to consumer i .

Interpretation: An allocation simply says who gets what.

Definition 10.1.5 (Feasible Allocation). The feasible set is the allocations that exhaust the aggregate endowment:

$$F(\mathbf{e}) = \left\{ \mathbf{x} : \sum_{i=1}^I x_k^i = \sum_{i=1}^I e_k^i \text{ for } k = 1, \dots, n \right\}.$$

With strongly monotone preferences, equilibrium allocations have no waste.

Definition 10.1.6 (General Equilibrium). A Walrasian general equilibrium is a price vector $\mathbf{p} \in \mathbb{R}_{++}^n$ and an allocation $\mathbf{x}$ such that:

1. each consumer maximizes utility subject to the budget determined by the value of their endowment,
2. supply equals demand in every market.

For each consumer i , utility maximization requires

$$u_i(\mathbf{x}^i) \geq u_i(\mathbf{y}^i)$$

for every bundle $\mathbf{y}^i$ satisfying

$$\mathbf{p} \cdot \mathbf{y}^i \leq \mathbf{p} \cdot \mathbf{e}^i.$$

Market clearing requires, for each good k ,

$$\sum_{i=1}^I x_k^i = \sum_{i=1}^I e_k^i.$$

General equilibrium is a big system of accounting equations plus individual optimization. Everyone solves their own problem taking prices as given. Prices are such that all markets clear simultaneously.

10.2 Aggregate Excess Demand

Assumption 5.1. For the main results, assume each consumer's preferences are continuous, strongly monotonic, and strictly convex on $\mathbb{R}_+^n$.

These are overly-strong assumptions, but they make our lives easier.

Theorem 10.2.1. *Under Assumption 5.1, for each consumer i and each price vector $\mathbf{p} \gg 0$, there is a unique optimal bundle. Each demand $x_k^i(\mathbf{p}, \mathbf{p} \cdot \mathbf{e}^i)$ is continuous in prices.*

Definition 10.2.2. For each good k , define aggregate excess demand by

$$z_k(\mathbf{p}) = \sum_{i=1}^I x_k^i(\mathbf{p}, \mathbf{p} \cdot \mathbf{e}^i) - \sum_{i=1}^I e_k^i.$$

We also define:

- aggregate demand x_k ,
- aggregate endowment e_k ,
- the excess-demand vector $\mathbf{z}(\mathbf{p}) = (z_1(\mathbf{p}), \dots, z_n(\mathbf{p}))$.

Theorem 10.2.3 (Properties of Excess Demand). *Under Assumption 5.1, for all $\mathbf{p} \gg 0$:*

1. *each $z_k(\mathbf{p})$ is continuous,*
2. *excess demand is homogeneous of degree zero:*

$$\mathbf{z}(t\mathbf{p}) = \mathbf{z}(\mathbf{p}) \quad \text{for all } t > 0,$$

3. *Walras' law holds:*

$$\mathbf{p} \cdot \mathbf{z}(\mathbf{p}) = 0.$$

Proof from Individual Budgets. Strong monotonicity implies each consumer exhausts their budget:

$$\sum_{k=1}^n p_k x_k^i(\mathbf{p}, \mathbf{p} \cdot \mathbf{e}^i) = \mathbf{p} \cdot \mathbf{e}^i = \sum_{k=1}^n p_k e_k^i.$$

Equivalently, each consumer's net trades have zero value:

$$\sum_{k=1}^n p_k [x_k^i(\mathbf{p}, \mathbf{p} \cdot \mathbf{e}^i) - e_k^i] = 0.$$

If every individual buys and sells goods of equal value, the economy as a whole must also buy and sell goods of equal value. □

These should feel conceptually familiar:

- Continuity: sums of continuous demands remain continuous.
- Homogeneity: demands are homogeneous of degree zero in prices and income.

- Walras' law: an accounting identity when each consumer is on the budget line.

In a two-good economy with $p_1, p_2 > 0$, Walras' law implies

$$p_1 z_1(\mathbf{p}) + p_2 z_2(\mathbf{p}) = 0,$$

so

$$p_1 z_1(\mathbf{p}) = -p_2 z_2(\mathbf{p}).$$

If $z_1(\mathbf{p}) > 0$, then $z_2(\mathbf{p}) < 0$. Excess demand in one market implies excess supply in the other.

If $z_1(\mathbf{p}) = 0$, then $z_2(\mathbf{p}) = 0$.

More generally, Walras' law also gives a useful n -market implication:

$$\sum_{k=1}^n p_k z_k(\mathbf{p}) = 0.$$

If $z_1(\mathbf{p}) = \dots = z_{n-1}(\mathbf{p}) = 0$, then $p_n z_n(\mathbf{p}) = 0$.

Since $p_n > 0$, the remaining market must also clear: $z_n(\mathbf{p}) = 0$. To find an equilibrium, we can drop one market equation.

More generally, excess demand somewhere must be matched by equal-value excess supply somewhere else.

Homogeneity of degree zero implies:

- only relative prices matter,
- we may normalize one price, for example by setting $p_1 = 1$.

This is the practical reason we solve general equilibrium problems “up to relative prices” rather than trying to determine an absolute price level.

10.3 Existence of General Equilibrium

Recall. A pair $(\mathbf{x}, \mathbf{p})$ is a general equilibrium if and only if:

- each consumer's allocation is utility maximizing given prices and endowment,
- excess demand is zero in each market.

Theorem 10.3.1 (Existence). *A general equilibrium exists whenever:*

- Assumption 5.1 holds,
- the total endowment of each good is strictly positive.

Interpretation: Existence is quite prevalent under standard assumptions.

The theorem gives a sufficient condition for existence. Equilibrium may still exist under weaker assumptions. If the assumptions fail, the only way to know is to check the specific economy.

Example 10.3.2. Now add John with utility

$$u_2(x_1, x_2) = x_1 x_2$$

and endowment

$$\mathbf{e}^2 = (2, 3).$$

Then John's demands are

$$x_1^2 = \frac{m^2}{2p_1}, \quad x_2^2 = \frac{m^2}{2p_2},$$

with income

$$m^2 = \mathbf{p} \cdot \mathbf{e}^2 = 2p_1 + 3p_2.$$

Substituting John's endowment income gives

$$x_1^2(\mathbf{p}) = \frac{2p_1 + 3p_2}{2p_1}, \quad x_2^2(\mathbf{p}) = \frac{2p_1 + 3p_2}{2p_2}.$$

Sarah's demands remain

$$x_1^1(\mathbf{p}) = \frac{2(2p_1 + 2p_2)}{3p_1}, \quad x_2^1(\mathbf{p}) = \frac{2p_1 + 2p_2}{3p_2}.$$

Because homogeneity and Walras' law hold: $p_1 = 1$, and we focus on just one market-clearing condition.

In market 1, total demand must equal total supply:

$$x_1^1(\mathbf{p}) + x_1^2(\mathbf{p}) = 2 + 2 = 4.$$

With $p_1 = 1$, market 1 clearing becomes

$$\frac{4(1 + p_2)}{3} + \frac{2 + 3p_2}{2} = 4.$$

Solving gives

$$p_2 = \frac{10}{17}.$$

So the equilibrium price vector is proportional to

$$\mathbf{p} = \left(1, \frac{10}{17}\right).$$

Using the equilibrium price vector:

$$\mathbf{x}^1 = \left(\frac{36}{17}, \frac{9}{5}\right), \quad \mathbf{x}^2 = \left(\frac{32}{17}, \frac{16}{5}\right).$$

Or,

$$\mathbf{x}^1 \approx (2.12, 1.8), \quad \mathbf{x}^2 \approx (1.88, 3.2).$$

Once equilibrium is found in market 1, Walras' law gives equilibrium in market 2 automatically.

Once one consumer's equilibrium bundle is found, the other consumer receives the residual feasible bundle.

It is still good practice to double check that each bundle is individually demanded at those prices.

Example 10.3.3. Suppose we have three consumers with

$$u_i(\mathbf{x}^i) = x_1^i x_2^i \quad \text{for } i = 1, 2, 3.$$

Endowments are

$$\mathbf{e}^1 = (19, 1), \quad \mathbf{e}^2 = (1, 19), \quad \mathbf{e}^3 = (10, 10).$$

Aggregate endowment is therefore

$$\mathbf{e}^1 + \mathbf{e}^2 + \mathbf{e}^3 = (30, 30).$$

Only relative prices matter, so normalize $p_2 = 1$. Then the value of aggregate endowment is

$$\mathbf{p} \cdot \sum_{i=1}^3 \mathbf{e}^i = p_1 \cdot 30 + 1 \cdot 30 = 30p_1 + 30.$$

For Cobb-Douglas utility $u_i(\mathbf{x}^i) = x_1^i x_2^i$, each consumer spends half of income on each good.

$$x_1^i(\mathbf{p}) = \frac{m^i}{2p_1}, \quad x_2^i(\mathbf{p}) = \frac{m^i}{2}, \quad m^i = \mathbf{p} \cdot \mathbf{e}^i.$$

The individual incomes are

$$m^1 = 19p_1 + 1, \quad m^2 = p_1 + 19, \quad m^3 = 10p_1 + 10.$$

Aggregate demand for good 1 is

$$\sum_{i=1}^3 x_1^i(\mathbf{p}) = \sum_{i=1}^3 \frac{m^i}{2p_1} = \frac{m^1 + m^2 + m^3}{2p_1}.$$

Using $m^1 + m^2 + m^3 = 30p_1 + 30$, we get

$$\sum_{i=1}^3 x_1^i(\mathbf{p}) = \frac{30p_1 + 30}{2p_1} = 15 + \frac{15}{p_1}.$$

Market clearing in good 1 requires this to equal aggregate supply 30:

$$15 + \frac{15}{p_1} = 30.$$

The good-1 market-clearing equation is

$$15 + \frac{15}{p_1} = 30.$$

Solving gives

$$p_1 = 1.$$

Since we chose the normalization $p_2 = 1$, one representative equilibrium price vector is

$$\mathbf{p} = (1, 1).$$

Without normalization, the equilibrium price ray is

$$\mathbf{p} = \lambda(1, 1) \quad \text{for any } \lambda > 0.$$

At $\mathbf{p} = (1, 1)$, each consumer has income

$$m^i = \mathbf{p} \cdot \mathbf{e}^i = 20.$$

Therefore,

$$x_1^i(\mathbf{p}) = \frac{m^i}{2p_1} = \frac{20}{2} = 10,$$

and

$$x_2^i(\mathbf{p}) = \frac{m^i}{2p_2} = \frac{20}{2} = 10.$$

So every consumer demands the same bundle:

$$\mathbf{x}^i(\mathbf{p}) = (10, 10).$$

Aggregate demand equals aggregate endowment:

$$\sum_{i=1}^3 \mathbf{x}^i(\mathbf{p}) = (10, 10) + (10, 10) + (10, 10) = (30, 30) = \sum_{i=1}^3 \mathbf{e}^i = (30, 30).$$

Therefore, $\mathbf{p} = (1, 1)$ is a Walrasian equilibrium price vector, with allocation

$$\mathbf{x}(\mathbf{p}) = ((10, 10), (10, 10), (10, 10)).$$

Good 1 was used to solve for the unknown relative price.

Good 2 clears by Walras' law, and the equality above verifies it directly.

10.4 Multiplicity and Failure

Important warning:

The Sarah example earlier did not satisfy all the assumptions.

Sarah's utility was

$$u_1(x_1, x_2) = x_1^2 x_2.$$

This utility is continuous, but the preferences do not satisfy all of Assumption 5.1 on the whole space $\mathbb{R}_+^n$.

In particular, on the boundary, if $x_2 = 0$, increasing x_1 does not raise utility. Also, bundles such as $(1, 0)$ and $(2, 0)$ give the same utility, and mixtures of them also give the same utility. So the assumptions of strong monotonicity and strict convexity on $\mathbb{R}_+^n$ fail.

Yet equilibrium still existed. Existence theorems are powerful, but they are not the whole story. Violations of the assumptions can sometimes lead to no competitive equilibrium: the model does not close. They can also lead to multiple general equilibria: different price systems and allocations may all satisfy the model.

Example 10.4.1 (No Equilibrium). Lisa has utility

$$u_1(x_1, x_2) = \max\{x_1, x_2\}$$

and endowment $\mathbf{e}^1 = (1, 2)$.

Jeff has utility

$$u_2(x_1, x_2) = \min\{x_1, x_2\}$$

and endowment $\mathbf{e}^2 = (2, 1)$.

The example violates the standard assumptions used in the existence theorem, so equilibrium need not exist.

Fix $p_1 = 1$. Then Lisa's demand for good 1 is piecewise:

$$x_1^1 = \begin{cases} 1 + 2p_2 & \text{if } p_2 > 1, \\ 0 \text{ or } 3 & \text{if } p_2 = 1, \\ 0 & \text{if } p_2 < 1. \end{cases}$$

Jeff's demand for good 1 is

$$x_1^2 = \frac{2 + p_2}{1 + p_2}.$$

Total supply of good 1 is 3. Check the cases one by one.

1. If $p_2 > 1$, demand is

$$1 + 2p_2 + \frac{2 + p_2}{1 + p_2} > 3.$$

2. If $p_2 = 1$, demand is either

$$0 + \frac{3}{2} = 1.5 < 3$$

or

$$3 + \frac{3}{2} = 4.5 > 3.$$

3. If $p_2 < 1$, total demand for good 1 is

$$0 + \frac{2 + p_2}{1 + p_2} < 2 < 3.$$

Thus total demand is never exactly equal to total supply.

That exhausts all possible values of p_2 . No equilibrium exists.

Example 10.4.2 (Multiple Equilibria Example). Now suppose Lisa has utility

$$u_1(x_1, x_2) = x_1 + 3.5x_2 - \frac{1}{2}x_2^2$$

and endowment $\mathbf{e}^1 = (4, 0)$.

Jeff has utility

$$u_2(x_1, x_2) = 3.5x_1 + x_2 - \frac{1}{2}x_1^2$$

and endowment $\mathbf{e}^2 = (0, 4)$.

These are quasi-linear utilities, but with respect to different goods.

Set $p_1 = 1$.

Then Lisa's income is $m^1 = 4$. Jeff's income is $m^2 = 4p_2$.

Lisa solves

$$\max_{x_1, x_2} x_1 + 3.5x_2 - \frac{1}{2}x_2^2 \quad \text{s.t.} \quad x_1 + p_2x_2 = 4.$$

Substituting $x_1 = 4 - p_2x_2$,

$$u_1 = 4 + (3.5 - p_2)x_2 - \frac{1}{2}x_2^2.$$

The first-order condition gives

$$x_2^1 = 3.5 - p_2.$$

Then

$$x_1^1 = 4 - p_2(3.5 - p_2) = 4 - 3.5p_2 + p_2^2.$$

Jeff solves

$$\max_{x_1, x_2} 3.5x_1 + x_2 - \frac{1}{2}x_1^2 \quad \text{s.t.} \quad x_1 + p_2x_2 = 4p_2.$$

Substituting $x_2 = 4 - \frac{x_1}{p_2}$,

$$u_2 = 4 + \left(3.5 - \frac{1}{p_2}\right)x_1 - \frac{1}{2}x_1^2.$$

The first-order condition gives

$$x_1^2 = 3.5 - \frac{1}{p_2}.$$

Then

$$x_2^2 = 4 - \frac{3.5}{p_2} + \frac{1}{p_2^2}.$$

Total supply of good 2 is 4. Demand is

$$x_2^1 + x_2^2 = (3.5 - p_2) + \left(4 - \frac{3.5}{p_2} + \frac{1}{p_2^2}\right).$$

Market clearing requires

$$3.5 - p_2 + 4 - \frac{3.5}{p_2} + \frac{1}{p_2^2} = 4.$$

Multiplying by $p_2^2 > 0$ gives

$$-p_2^3 + 3.5p_2^2 - 3.5p_2 + 1 = 0,$$

or

$$\left(p_2 - \frac{1}{2}\right)(p_2 - 1)(p_2 - 2) = 0.$$

The market-clearing roots are

$$p_2 = \frac{1}{2}, \quad 1, \quad 2.$$

For each root, good 2 clears by construction. Good 1 also clears by Walras' law.

$\mathbf{p} = (1, p_2)$	Lisa $\mathbf{x}^1$	Jeff $\mathbf{x}^2$	$u_1(\mathbf{x}^1)$	$u_2(\mathbf{x}^2)$
$(1, \frac{1}{2})$	(2.5, 3)	(1.5, 1)	8.5	5.125
(1, 1)	(1.5, 2.5)	(2.5, 1.5)	7.125	7.125
(1, 2)	(1, 1.5)	(3, 2.5)	5.125	8.5

Lisa prefers the low p_2 equilibrium:

$$\left(1, \frac{1}{2}\right) \succ_1 (1, 1) \succ_1 (1, 2).$$

Jeff prefers the high p_2 equilibrium:

$$(1, 2) \succ_2 (1, 1) \succ_2 \left(1, \frac{1}{2}\right).$$

Lesson from the Counterexamples:

- Competitive equilibrium need not be unique.
- Competitive equilibrium need not even exist.
- Standard assumptions are useful because they rule out these pathologies.
- Once those assumptions are weakened, the model has to be checked economy by economy.

Source note: The multiple-equilibria example is from Bergstrom, Shimomura and Yamamoto, simplifying an earlier Shapley-Shubik construction.

11 Welfare

11.1 Pareto Efficient

Definition 11.1.1 (Pareto Efficient Allocation). An allocation $\mathbf{x} \in F(\mathbf{e})$ is Pareto efficient in an economy E if:

- $\mathbf{x}$ is feasible, and
- there is no other allocation $\mathbf{y} \in F(\mathbf{e})$ such that every consumer is at least as well off at $\mathbf{y}$ as at $\mathbf{x}$, with one consumer strictly better off. Mathematically,

$$u_i(\mathbf{y}^i) \geq u_i(\mathbf{x}^i) \quad \text{for all } i, \quad u_j(\mathbf{y}^j) > u_j(\mathbf{x}^j) \quad \text{for some } j.$$

It is the standard benchmark in economics. It rules out avoidable waste and mutually beneficial trades. But it says nothing about fairness or equality. Very unequal allocations can still be Pareto efficient.

Interpretation: Pareto efficiency is a starting point for welfare analysis, not the end of it.

The definition is asymmetric in practice:

- To show an allocation is not Pareto efficient, it suffices to find one feasible reallocation that makes everyone weakly better off and someone strictly better off.
- To show an allocation is Pareto efficient, you must rule out all such feasible reallocations.

Example 11.1.2. Consider a two-person exchange economy. Anne has utility $u_A(\mathbf{x}^A) = x_1^A x_2^A$ and endowment $\mathbf{e}^A = (5, 1)$. Brad has utility $u_B(\mathbf{x}^B) = (x_1^B)^2 x_2^B$ and endowment $\mathbf{e}^B = (5, 4)$.

The initial allocation is

$$(\mathbf{e}^A, \mathbf{e}^B) = ((5, 1), (5, 4)).$$

It is feasible, so the only question is whether some other feasible allocation makes both weakly better off and one strictly better off. Brad values good 1 relatively more than Anne does and has lots of good 2 to trade, so probably yes.

Try the allocation

$$\mathbf{x}^A = (4, 2), \quad \mathbf{x}^B = (6, 3).$$

This is feasible because total goods remain

$$x_1^A + x_1^B = 10, \quad x_2^A + x_2^B = 5.$$

Utilities become

$$u_A(4, 2) = 8 > u_A(5, 1) = 5, \quad u_B(6, 3) = 6^2 \cdot 3 = 108 > u_B(5, 4) = 100.$$

So the endowment allocation is not Pareto efficient.

The goal is to identify the set of Pareto efficient feasible allocations. For this two-person exchange economy, feasible allocations satisfy

$$F(\mathbf{e}) = \{(\mathbf{x}^A, \mathbf{x}^B) \in \mathbb{R}_+^2 \times \mathbb{R}_+^2 : x_1^A + x_1^B = 10, x_2^A + x_2^B = 5\}.$$

Each person gets nonnegative amounts of both goods. This set is convex.

- Observation 1: No waste at an efficient allocation. Because both utility functions are strictly increasing, every Pareto efficient allocation must use all available goods.

$$x_1^A + x_1^B = 10, \quad x_2^A + x_2^B = 5.$$

- Observation 2: MRS equalization. At an interior Pareto efficient allocation with smooth strictly convex preferences, the consumers' marginal rates of substitution must be equal.

$$MRS_{2 \text{ for } 1}^A = MRS_{2 \text{ for } 1}^B.$$

If the MRS values differ, a small trade can make both consumers better off.

Definition 11.1.3. For a utility function $u(x_1, x_2)$,

$$MRS_{2 \text{ for } 1} = \frac{MU_1}{MU_2}.$$

It measures how much of good 2 the person is willing to give up for one more unit of good 1.

For Anne,

$$u_A(\mathbf{x}^A) = x_1^A x_2^A \Rightarrow u_1^A = x_2^A, \quad u_2^A = x_1^A, \quad MRS^A = \frac{x_2^A}{x_1^A}.$$

At (5, 1),

$$MRS^A = \frac{1}{5}.$$

For Brad,

$$u_B(\mathbf{x}^B) = (x_1^B)^2 x_2^B \Rightarrow u_1^B = 2x_1^B x_2^B, \quad u_2^B = (x_1^B)^2, \quad MRS^B = \frac{2x_2^B}{x_1^B}.$$

At (5, 4),

$$MRS^B = \frac{8}{5}.$$

At the endowment,

$$MRS^A = \frac{1}{5} < \frac{8}{5} = MRS^B.$$

So Brad is willing to give up more of good 2 for an extra unit of good 1 than Anne requires to give up that unit.

- Brad values good 1 more intensely at the margin.
- Anne values good 2 more intensely at the margin.
- This gap creates gains from trade.

Pareto efficient allocations must satisfy:

$$x_1^A + x_1^B = 10, \quad x_2^A + x_2^B = 5, \quad \frac{x_2^A}{x_1^A} = \frac{2x_2^B}{x_1^B}.$$

We solve these three equations for the efficient set.

From no waste,

$$x_1^B = 10 - x_1^A, \quad x_2^B = 5 - x_2^A.$$

Substitute into the MRS condition:

$$\frac{x_2^A}{x_1^A} = \frac{2(5 - x_2^A)}{10 - x_1^A}.$$

Rearranging yields

$$10x_2^A + x_1^A x_2^A - 10x_1^A = 0.$$

Solving the efficiency equation for x_2^A gives

$$x_2^A(x_1^A) = \frac{10x_1^A}{10 + x_1^A}, \quad 0 < x_1^A < 10.$$

Then Brad gets the residual bundle:

$$x_1^B = 10 - x_1^A, \quad x_2^B = 5 - x_2^A(x_1^A).$$

So the Pareto set is parameterized by Anne's choice of x_1^A .

We can write the efficient set as

$$PE = \left\{ (\mathbf{x}^A, \mathbf{x}^B) : 0 \leq x_1^A \leq 10, x_2^A = \frac{10x_1^A}{10 + x_1^A}, x_1^B = 10 - x_1^A, x_2^B = 5 - x_2^A \right\}.$$

- Every point on this curve is Pareto efficient.
- Every Pareto efficient allocation in this example has this form.

The Pareto set includes the extremes:

$$((10, 5), (0, 0)), \quad \text{and} \quad ((0, 0), (10, 5)).$$

- Best for Anne, worst for Brad.
- Best for Brad, worst for Anne.

Important warning: Pareto efficiency does not mean “fair” or “balanced”.

Finding the full Pareto set is not always simple. If all utility functions are concave in addition to other standard conditions, one method is to maximize a weighted sum of utilities and vary the weights. See Propositions 16.E.1 and 16.F.1 of Mas-Colell, Whinston, and Green.

$$\max_{\mathbf{x}} \alpha u_1(\mathbf{x}^1) + (1 - \alpha)u_2(\mathbf{x}^2).$$

You could use this method with a different concave representation of Anne and Brad’s preferences, but no-waste plus MRS equalisation is often faster.

11.2 First Fundamental Theorem

Theorem 11.2.1 (First Fundamental Theorem of Welfare Economics). *In an exchange economy, if every person’s utility function is strictly increasing on $\mathbb{R}_+^n$, every Walrasian equilibrium allocation is Pareto efficient.*

Proof. The key proof intuition mirrors Walras’ law. At equilibrium prices, each consumer chooses the best affordable bundle. If there were a feasible allocation making everyone weakly better off and someone strictly better off, its total value would have to exceed total endowment value. But feasibility prevents that. $\square$

Competitive equilibrium implies efficiency. This is the main normative justification for competitive markets in the model. Not much is needed:

- Consumers must choose bundles on their budget lines.
- Monotonicity is enough to push demand to the budget line.
- If equilibrium does not exist, the theorem says nothing.

The theorem does not say the following:

- It does not say every Pareto efficient allocation is a competitive equilibrium.
- It does not say the equilibrium is fair.
- It does not say equilibrium exists in every economy.

11.3 The Core

Pareto efficiency asks whether the whole economy could reallocate and improve. The core asks a stronger question: could some subgroup leave and do better using only its own endowments? If yes, the proposed allocation is unstable against coalition deviations.

Definition 11.3.1 (Blocking). A subset of consumers $S \subseteq \{1, \dots, I\}$ blocks an allocation $\mathbf{x} \in F(\mathbf{e})$ if there exists another allocation $\mathbf{y}$ such that:

1. It is feasible for the coalition,

$$\sum_{i \in S} \mathbf{y}^i = \sum_{i \in S} \mathbf{e}^i.$$

2. Every member of the coalition is weakly better off,

$$u_i(\mathbf{y}^i) \geq u_i(\mathbf{x}^i) \quad \text{for all } i \in S,$$

with at least one strict inequality.

Definition 11.3.2 (Core). The core of an exchange economy with endowment $\mathbf{e}$, denoted $C(\mathbf{e})$, is the set of unblocked feasible allocations.

- No individual can leave and do better alone.
- No pair can leave and do better together.
- No larger coalition can leave and do better together.

The core is a central concept in cooperative game theory, in which players are allowed to cooperate, form coalitions, and split surplus. An allocation is acceptable only if no coalition can propose another feasible split that all coalition members weakly prefer and one strictly prefers.

If the grand coalition $\{1, \dots, I\}$ can block an allocation, then that allocation is not Pareto efficient. So every allocation in the core must be Pareto efficient. But not every Pareto efficient allocation is necessarily in the core.

Proposition 11.3.3.

$$C(\mathbf{e}) \subseteq \text{Pareto efficient set.}$$

For a two-person pure exchange economy, the requirement is simple:

- Start from the Pareto set.
- Then require that each person gets at least the utility from his or her endowment.

With only two consumers, if neither individual wants to walk away, then no smaller coalition remains to check.

For the Anne-Brad economy, the Pareto set is

$$PE = \left\{ (\mathbf{x}^A, \mathbf{x}^B) : 0 \leq x_1^A \leq 10, x_2^A = \frac{10x_1^A}{10 + x_1^A}, x_1^B = 10 - x_1^A, x_2^B = 5 - x_2^A \right\}.$$

To find the core, impose the condition that each person weakly prefers the allocation to his or her own endowment.

Anne's endowment utility is

$$u_A(5, 1) = 5.$$

Along the Pareto set,

$$u_A(\mathbf{x}^A) = x_1^A x_2^A = x_1^A \cdot \frac{10x_1^A}{10 + x_1^A}.$$

So Anne does not block if

$$x_1^A \cdot \frac{10x_1^A}{10 + x_1^A} \geq 5.$$

This simplifies to

$$(x_1^A)^2 - 0.5x_1^A - 5 \geq 0,$$

which requires

$$x_1^A \geq 2.5.$$

Brad's endowment utility is

$$u_B(5, 4) = 100.$$

The old lecture rewrote his condition in terms of x_1^B :

$$u_B(\mathbf{x}^B) = (x_1^B)^2 x_2^B = (x_1^B)^2 \cdot \frac{5x_1^B}{20 - x_1^B}.$$

So Brad does not block if

$$(x_1^B)^3 \geq 20(20 - x_1^B),$$

or equivalently

$$(x_1^B)^3 + 20x_1^B - 400 \geq 0.$$

This requires approximately

$$x_1^B \geq 6.47.$$

Because total good 1 is fixed,

$$x_1^B = 10 - x_1^A.$$

So Brad's no-blocking condition becomes

$$10 - x_1^A \geq 6.47 \Rightarrow x_1^A \leq 3.53.$$

Combining both individual no-blocking constraints gives the core interval for Anne's efficient good-1 consumption.

Altogether the core is:

$$C(\mathbf{e}) = \left\{ (\mathbf{x}^A, \mathbf{x}^B) : 2.5 \leq x_1^A \leq 3.53, x_2^A = \frac{10x_1^A}{10 + x_1^A}, x_1^B = 10 - x_1^A, x_2^B = 5 - x_2^A \right\}.$$

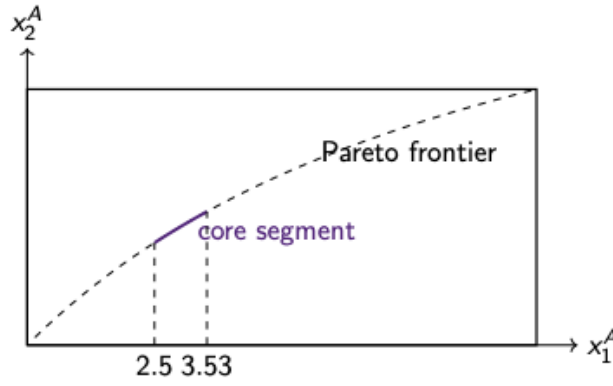


Figure 6: The core is a strict subset of the Pareto set.

Extreme efficient allocations are excluded because one of the individuals would walk away.

Solve for competitive equilibrium. Fix $p_1 = 1$ and solve for market clearing.

Anne's demand for good 1 is

$$x_1^A = \frac{5 + p_2}{2}.$$

Brad's demand for good 1 is

$$x_1^B = \frac{2(5 + 4p_2)}{3}.$$

Market clearing for good 1 requires

$$\frac{5 + p}{2} + \frac{2(5 + 4p)}{3} = 10 \Rightarrow p = \frac{25}{19} \approx 1.32.$$

Here p denotes the equilibrium value of p_2 .

At that price,

$$x_1^A = \frac{5 + p}{2} = \frac{60}{19} \approx 3.16.$$

This lies inside the core interval:

$$2.5 \leq x_1^A \leq 3.53.$$

The competitive equilibrium allocation is Pareto efficient by the first welfare theorem. It is also stable against coalition deviations in this example. So it lands inside the core rather than merely on the full Pareto set.

Theorem 11.3.4 (Core Property of General Equilibrium). *If each u_i is strictly increasing on $\mathbb{R}_+^n$, then every Walrasian equilibrium allocation is in the core:*

$$W(\mathbf{e}) \subseteq C(\mathbf{e}).$$

Proof. The proof logic is similar to the first welfare theorem: prices support the allocation. $\square$

Competitive equilibrium is not just Pareto efficient. No coalition can profitably break away and reallocate its own endowment.

If an allocation came from competitive equilibrium, nobody wants to leave the market arrangement and form a smaller trading group. This gives a cooperative-stability interpretation of equilibrium. Similar caveats: if competitive equilibrium does not exist, the theorem says nothing. The core is usually a large set; competitive equilibrium is a smaller solution concept.

Things are more complicated with more than two consumers.

Example 11.3.5. Suppose three consumers have $u_i(\mathbf{x}^i) = x_1^i x_2^i$. Endowments are

$$\mathbf{e}^1 = (19, 1), \quad \mathbf{e}^2 = (1, 19), \quad \mathbf{e}^3 = (10, 10).$$

Their endowment utilities are 19, 19, 100.

Consider the allocation

$$\mathbf{x}^1 = (9, 9), \quad \mathbf{x}^2 = (9, 9), \quad \mathbf{x}^3 = (12, 12).$$

It is Pareto efficient and makes each consumer better off than at the endowment:

$$81 > 19, \quad 81 > 19, \quad 144 > 100.$$

But the first two consumers can block it by reallocating their joint endowments to

$$(10, 10), \quad (10, 10).$$

So this allocation is not in the core.

With more than two consumers, checking only individual deviations is not enough. Coalitions of size two or more can block an allocation. So the core becomes substantially harder to compute. We have to check all relevant subgroups.

For the three-consumer economy with identical utilities $u_i(\mathbf{x}^i) = x_1^i x_2^i$, the Pareto set is

$$PE = \left\{ (\mathbf{x}^1, \mathbf{x}^2, \mathbf{x}^3) : \mathbf{x}^1 = (a, a), \mathbf{x}^2 = (b, b), \mathbf{x}^3 = (c, c), a, b, c \geq 0, a + b + c = 30 \right\}.$$

Equal MRS implies each consumer must consume goods in a 1 : 1 ratio. No waste implies the common coordinates must sum to total endowment.

For consumers 1 and 2 not to leave, we need

$$a, b \geq \sqrt{19} \approx 4.36,$$

and for consumer 3, we need $c \geq 10$.

Coalition $\{1, 2\}$ can block whenever $a + b < 20$. In that case,

$$a' = a + \frac{20 - a - b}{2}$$

and

$$b' = b + \frac{20 - a - b}{2}$$

is feasible and better.

This, plus $c \geq 10$, plus total endowment $a + b + c = 30$, means $a + b = 20$ and $c = 10$ in the core.

Fix consumer 1's utility at some level u . Let coalition $\{1, 3\}$ reallocate its joint endowment $(29, 11)$ to maximize consumer 3's utility subject to consumer 1 receiving utility u .

That is,

$$\max(29 - x_1)(11 - x_2) = (29 - x_1) \left(11 - \frac{u}{x_1}\right) \quad \text{s.t.} \quad u = x_1 x_2.$$

This yields:

$$x_1 = \sqrt{\frac{29u}{11}}, \quad x_2 = \sqrt{\frac{11u}{29}}.$$

If consumer 3 gets utility 100, this block will be an improvement whenever

$$(29 - \sqrt{\frac{29a^2}{11}})(11 - \sqrt{\frac{11a^2}{29}}) = (\sqrt{319} - \sqrt{a^2})^2 > 10^2 \quad \Rightarrow \quad a < \sqrt{319} - 10 \approx 7.86.$$

So, by symmetry (for coalition $\{2, 3\}$), the core must have $a, b \geq 7.86$.

Altogether,

$$C(\mathbf{e}) = \{(\mathbf{x}^1, \mathbf{x}^2, \mathbf{x}^3) : \mathbf{x}^1 = (a, a), \mathbf{x}^2 = (b, b), \mathbf{x}^3 = (c, c)\},$$

where

$$a, b \geq 7.86, \quad c = 10, \quad a + b + c = 30.$$

Recall from last lecture that the competitive equilibrium gives each consumer $(10, 10)$.

11.4 Second Fundamental Theorem

Theorem 11.4.1 (Second Fundamental Theorem of Welfare Economics). *Consider an exchange economy E in which Assumption 5.1 is satisfied and the total endowment of each good is greater than zero. Every Pareto efficient allocation $\mathbf{x}$ can be obtained by appropriately redistributing the endowments of the consumers.*

The theorem says: start from a Pareto efficient allocation you want, redistribute endowments appropriately, then the competitive market can support that allocation as an equilibrium. Efficiency and distribution can, in principle, be separated.

The exchange-economy construction is simple:

- Let $\mathbf{e}$ be the original endowment vector.
- Let $\mathbf{x}$ be the desired Pareto efficient allocation.
- Choose a new endowment profile $\mathbf{e}'$ such that $\mathbf{e}' = \mathbf{x}$.
- Under the theorem's assumptions, the new endowment is supported as a general equilibrium.

There are usually many Pareto efficient allocations. A society may prefer one efficient allocation to another on distributional grounds. The second welfare theorem says markets can still decentralize that chosen efficient allocation after redistribution.

In richer production economies, the idea becomes: tax and transfer wealth or income first. Then let markets work from that redistributed starting point. The market mechanism can still deliver the targeted efficient allocation if the assumptions hold.

Compared with the first theorem, the second theorem needs more structure.

- existence of competitive equilibrium,
- convexity-type assumptions behind Assumption 5.1,
- positive total endowments.

To support an arbitrary Pareto efficient allocation with prices, we need enough convexity and continuity to separate supporting hyperplanes from the feasible set.

11.5 Extensions and Limitations

Both welfare theorems extend to production economies. The second theorem is less trivial there and needs conditions on production technologies.

- Increasing returns can prevent competitive equilibrium from existing.
- Then the welfare theorems become vacuous: there is no competitive equilibrium to evaluate.
- Alternative solution concepts, such as Nash equilibrium, may become more appropriate.

This is the formal basis for the idea that taxes and transfers can guide an economy toward efficiency while leaving decentralized markets to do the allocation work.

Limitations of the First Welfare Theorem:

1. The first welfare theorem implicitly assumes no externalities.
 - Utility depends only on one's own bundle.
 - If one person's consumption or production affects another person's utility, competitive equilibrium need not be Pareto efficient.
2. The first welfare theorem also assumes that the price system conveys the relevant information.
 - If information is limited or asymmetric, equilibrium may not be efficient.
 - In such environments, noncompetitive institutions can sometimes get closer to efficient outcomes.

Limitations of the Second Welfare Theorem:

1. Governments cannot tax and transfer perfectly.
2. The theorem assumes enough knowledge of preferences to know which Pareto efficient allocation to target.
3. Incentive effects are ignored.